Abstract

Deepfakes has been prevalent and transformative in digital landscape through highly realistic synthetic media. Although it provides benefits on fields of entertainment, marketing, and education, there has been emerging concerns on privacy violations, reputation harm, and widespread of misinformation. This quantitative study investigates the interconnection between deepfake awareness, ethical position, and detection ability in influencing reporting behaviour of students. Survey data was collected from 60 students in Metro Manila to examine their awareness, understanding, and response to deepfake technology. It was found that higher levels of deepfake awareness profoundly increase the likelihood of reporting, with detection ability acting a partial mediating role. Moreover, the ethical position moderates this direct relationship, which implies that students with stronger moral judgement exhibit different levels of willingness to report deepfake content. The study emphasizes the importance of enhancing digital literacy programs, ethical education and implementing policy interventions to address societal risks of deepfake technology, while also promoting a responsible use of AI technology.

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From Awareness to Action: The Role of Ethical Position and Detection Ability in Deepfake Awareness and Reporting Behavior among Students in Metro Manila

Introduction

Background of the Study

In recent years, the advancement of Artificial Intelligence (AI) technology has become extremely relevant and sophisticated, and it will continue to be so in the succeeding years. It has had a wide and transformative impact on human life and industries. Despite its numerous applications, AI has been increasingly challenging and has raised global and ethical concerns, particularly in the area of Generative AI. One of these concerns is the recent surge of fake videos, images, and audio on the internet, known as deepfakes (Young, 2021). The term “deepfake” is a combination of “deep learning” and “fake” which refers to using DL neural network (NN) algorithms to fabricate media and appear to be real (Silva et al., 2022). A deepfake is a specific type of synthetic media where an individual’s video or image is swapped with another person’s likeness (Somers, 2020). Deepfake technology utilizes powerful Artificial Intelligence (AI) algorithms and computer graphics, making it difficult to distinguish real and fake media.

Gupta (2023) claims that deepfake has potential benefits for the entertainment, education, and marketing industries because of its accessibility and reduced workload of creating engaging content. However, it's important to point out the potential for deceit and disinformation, especially with the rise of content sharing on social media platforms, making it a dangerous tool for the dignity, consent, and privacy of affected individuals. Deepfake is represented as a double-edged sword in digital media, and Fallis (2020) states that the emergence of deepfakes presents a new epistemic threat. A key ethical concern is how accessible or easy it is for individuals to produce deepfakes, which raises issues about the creators’ responsibility and the lack of technological and legal systems to prevent the harm they cause. Sharma et al. (2023) found that many reputable political individuals have been victims of nonconsensual deepfake videos that have gone viral on social media platforms. Some of these individuals are President Obama, President Zelenskyy, and other politicians, all of whom have fallen victim to reputational damage and public distrust. This issue does not only concern famous individuals but also ordinary people, leading to serious social, psychological, and financial outcomes.

Research Problem and Aim

Fake news and content have been around for a long time and have been manipulating public opinions on various topics. The advancements in Artificial Intelligence (AI) and the internet have been a gateway to create new forms of manipulation and disinformation. The emergence of deepfakes amplifies this widespread issue in social media platforms due to its notable realistic content and its minimal expertise in dissemination. According to Moreno (2024), the victims of deepfakes may experience psychological trauma, financial loss, social isolation, reputational damage, and in some cases, loss of life.

There are plenty of studies that highlight the growing prevalence and harmful use of deepfakes. However, there were inadequacies in how well individuals recognize and report deepfake content in regard to deepfake awareness and ethical stance. Consequently, this has urged the researcher to investigate the levels of deepfake awareness, ethical position, and detection ability and how these factors affect the likelihood of an individual reporting deepfake content. In a study by Akter and Rhaman (2024), a notable percentage (12%) of young people have never heard of deepfakes, which means there's an existing knowledge gap that needs to be addressed. The study also explores whether the ability to detect deepfakes increases the likelihood of an individual reporting the content, since this area remains largely unexplored. The researcher desires to lessen the unethical exploitation of deepfake technology, promote awareness among individuals, and encourage them to report fake content. The study aims to contribute valuable insights into the public's awareness and moral principles surrounding deepfake content, with potential implications for ethics and digital literacy about deepfakes. It also aims to contribute to understanding the main factors that drive individuals to report deepfake content.

Research Questions

1. To what extent does the individual's detection ability mediate the relationship between deepfake awareness and their likelihood to report deepfake content?
2. How does an individual's ethical position moderate the strength or direction of the relationship between deepfake awareness and reporting behavior?
3. To what extent does an individual's deepfake awareness directly influence their likelihood to report deepfake content?

Research Objectives

1. To investigate whether and how detection ability serves as a mediator between deepfake awareness and an individual's likelihood to report deepfake content.
2. To determine how an individual's ethical position serves as a moderator between deepfake awareness and an individual's likelihood to report deepfake content.
3. To examine the direct effect of deepfake awareness on an individual's likelihood to report deepfake content.

Research Significance, Scope, and Delimitation

Research Significance

The study provides valuable information about the role of deepfake awareness, ethical position, and detection ability to influence the reporting behavior of individuals, which can significantly combat the spread of misinformation and mislead society. This study addresses the existing literature gaps and contributes knowledge regarding the student's ethical stance, deepfake awareness, detection skills, and reporting intention of individuals to manipulate media. This area is often overlooked, wherein existing studies predominantly focused on the technological areas, harmful effects, and psychological impact of deepfakes instead of its potential for prevention through awareness and behavioral response on this issue. An example is the study of Mehta et al. (2021), where they have proposed detection or recognition tools for deepfake videos. Despite the rapid progress and advancements of detection tools, deepfake detection still experiences many hurdles in accurately identifying fake media (Rana et al., 2022). Other studies, such as Gupta (223) and Sharma et al. (2023), emphasize the potential harm and emergence of deepfakes, however, there is little empirical research examining the individual's level of awareness and actions taken when encountering deepfakes. Also, there is little research that discusses the ethical position and responsibility of users towards consuming and sharing this kind of content. Knowing the ethical position of students on deepfakes helps the researcher understand how informed they are on the ethical dilemmas and associated risks of this issue. Therefore, this study will expand the knowledge for academic research, digital literacy, and ethics concerning the interconnection between awareness and proactive actions of individuals on this relevant issue.

Practically, the study provides insights into the ability of individuals to distinguish real and fake content, which is critical for the development of mitigation strategies and interventions to alleviate the widespread misinformation on social media platforms. An application of this is in further research for Akter and Rhaman's (2024) study, wherein they suggested launching educational campaigns like seminars and workshops in schools for the students to broaden their understanding of deepfakes better. The study's findings inform the public about improving digital literacy programs, which would guide educators and management to create seminars to carefully assess online content. This would aid students in becoming aware of the ethical implications amidst the advancements in Artificial Intelligence (AI) technology. This proactive approach makes students less vulnerable or victimized by misinformation in today's media-saturated digital world. Given that society in this digital age uses social media routinely, being a victim of misinformation and manipulation can lead to drastic consequences and controversy. Thus, this study inspires individuals to report malicious deepfake content to protect the integrity of content posted online. This study will also inform future research

whether its scope is understanding the public's awareness and reaction toward deepfakes.

Furthermore, the study can guide policymakers in implementing clear regulations that safeguard the dignity, rights, and privacy of individuals while also promoting responsible and ethical use of Generative AI. Social media platforms may enhance the reporting features and strengthen the mechanisms for removing and flagging malicious fake media. According to Lu (2020), it was found that there is a lack of regulations and punishment for the misuse of deepfake technology. By understanding the causal inferences between deepfake detection and reporting behavior, this study can supply information for the development of frameworks regarding the behavioral response of individuals to deepfake content. This study will shed light on societal awareness and add depth to how individuals handle the complexities of fake media.

Scope and Delimitations

The study primarily focuses on evaluating the level of awareness among individuals about deepfakes and analyzing how these varying levels of awareness influence their ability to detect deepfake content better. The study also aims to determine how an ethical position moderates the relationship between deepfake awareness and reporting behavior. In the conceptual framework section, there illustrates and explains the relationships between variables to show how one is connected to another. In doing so, the study will further explore the causality between the predictors and the reporting behavior of individuals. The demographics of the study are narrowed down to students within Metro Manila who are also active social media users, as the study concerns the prevalence of deepfake content spreading across social media platforms. Metro Manila is an ideal geographical context, as it is a centralized urban area and generally has a high concentration of internet or social media users in the Philippines. This is supported by the nationwide survey findings of Pulse Asia Research Inc. (2021), which indicate that 72% of people in Metro Manila have internet access compared to the national average of 63%. In addition, there is a high frequency of internet usage among Metro Manila residents (58%) compared to other regions in the Philippines. Given that there is a wide range of schools and universities within the locale, the insights gained from this study can help inform educational campaigns relating to this issue.

Students are chosen to be the respondents of the study as they are generally equipped with the digital skills needed to evaluate information online, as asserted by the study of Javier (2023). This choice also aims to address the knowledge gap among young adults regarding deepfakes, as emphasized in the study of Akter and Rhaman (2024). Students are also increasingly exposed to social media, which raises their vulnerability to digital threats that surface on these platforms (Abrams 2022; Office of the Surgeon General, 2023). This is a cross-sectional study that will start in November 2024 and is targeted to be completed by May 2025.

The study will exclude non-students from Metro Manila and mainly focus on students for participation. Additionally, it covers social media platforms and does not include other social networks. The researcher aims to investigate potential cause-and-effect relationships between the students' awareness, ethical positions, detection abilities, and reporting behaviours. However, the study will not undergo any experimental procedure; rather, it will rely on correlational analysis with a causal inference approach to evaluate these variables. The study will not address the technological aspects of deepfakes, such as face recognition algorithms and deep learning models. Finally, the researcher will employ a quantitative research design, which utilizes a research questionnaire and statistical analyses to achieve the study objectives.

Literature Review

Deepfake Awareness

"Awareness is the first defence against any cyber-security threat.". This statement by Andrew Bud (2019) justifies that awareness plays a critical role in fighting cyber threats like deepfakes. Deepfake awareness is defined as the acknowledgment and understanding of the consequences and existence of deepfake technology in different contexts. It highlights the necessity to be alert to the presence of deepfakes, especially on well-known domains such as politics and media representation. Interestingly, a nationwide survey conducted by iProov (2019) found that 72% of the UK public had never heard of deepfakes. Once informed, 65% of the respondents expressed a decline in their trust in the Internet, with 42% recognizing identity theft as their topmost concern. Additionally, it is forecasted that, by 2026, 90% of online content will be synthetically generated by Artificial Intelligence (IDCA Press, 2022). These reports emphasize the urgent need for increased awareness to defend against cyber threats like deepfakes.

A study conducted by Sippy et al. (2024) states that while awareness of deepfakes is growing, exposure to harmful deepfakes remains low. A national survey of 1,403 UK adults found that only 15% reported encountering deepfakes used for fraud and misinformation. However, 90.4% of individuals expressed high levels of concern about deepfakes spreading exploitation, manipulation, and distrust among the public. A study by Collard (2024) indicates that while there is a growing deepfake awareness, there is still a significant gap in detection ability. Collard (2024) argues that media literacy, along with public awareness, is critical for understanding how deepfakes circulate and the psychological tactics used by harmful creators. The studies imply that awareness and fears of deepfakes were significantly high, and the authors recommend addressing the need for media literacy to improve resilience and detection.

Blancaflor et al. (2023) further elaborated that deepfakes have gone beyond entertainment, wherein they could also be used for revenge porn and political tools. A survey of 165 Filipinos found a correlation between exposure to social media, awareness of deepfakes, and detection ability. In addition, those individuals with prior exposure recognized the lack of legislation on deepfakes in the Philippines. This is supported by the news article published by Felicia (2024), whereby the DICT Secretary Ivan John Uy called for comprehensive deepfake legislation in the Philippines. He emphasized that the existing laws encompassing scams, misinformation, and deepfakes are inadequate and fragmented. Uy cited the "online harm law" in Singapore as a model, which stresses the need for cohesive regulation beyond Article 154 of the Revised Penal Code. It can be concluded that the Philippines has severely limited understanding and solutions to mitigate deepfakes, and the urgent need for public action and legislation will help reduce its harmful impacts.

Ethical Position

Ethical position is defined as a person's stance on a particular issue, which involves their moral code that guides their behavior and decision-making (Baz, 2005). The study is particularly interested in the Ethics Position Theory, which suggests that every individual makes decision in their daily lives about what is right and wrong (Forsyth, 2019). It is said that individuals develop their unique moral beliefs, which are influenced by two moral concerns: idealism and relativism. Idealism is defined as whether an action causes or may cause harm to others. Relativism is defined as whether an action aligns with the general moral standards. The Ethics Position Theory is connected to two philosophical approaches to ethics: deontology and consequentialism (Forsyth, 2019).

To establish a connection between deepfakes and ethical position, Gomez (2024) examined deepfakes based on two ethical frameworks: consequentialism and deontology, which is closely related to the Ethics Position Theory. Consequentialism determines whether deepfakes are morally "good" based on their outcomes or impact on society. If the benefits outweigh the harm, then it is deemed to be good. Gomez (2024) provided clear evidence of how deepfakes are used in legal and societal contexts in harmful ways. For instance, Russia has created a deepfake video of the Ukrainian president declaring that Ukraine is surrendering during wartime. A 2019 UK custody case threatened the justice system by using deepfake technology as evidence during trials (Edwards, 2022). It was also found that 63% of U.S. adults said that deepfakes cause "a great deal of confusion" (Gottfried, 2019). While confusion does not directly harm people, it can still mislead individuals to make poor decisions that result in harmful outcomes. Hence, consequentialism itself won't determine if deepfakes are ethical or not due to the oversight of ethical factors. 96% of all deepfakes exist to be deepfake porn (Houser, 2019). If an individual creates a deepfake porn privately, no one would be harmed, which from the consequentialist perspective, would not be seen as unethical. However, the creation of non-consensual deepfake porn is harmful and illegal, which underscores the importance of having another ethical viewpoint.

Another ethical framework is deontology. Deontology views actions to be inherently right or wrong, regardless of the consequences or outcomes of the action. In the application of deepfakes, morality is based on whether it is deceptive or not. Deepfakes are known to create deception and violate moral duties by distorting reality through fake content. Immanuel Kant, who centers the deontological theory, believes that deception is inherently wrong. A deepfake will only be ethical if it is consented to by all parties and transparency that the media is fake. Deepfakes should only be utilized for legitimate purposes like lawful or academic research (Moehring, 2024). For instance, deepfakes can be ethical when used to aid individuals with disabilities in communication. Therefore, in the context of deontology, deepfakes that are created to be deceptive and spread falsehoods are immoral.

Yang (2024) revealed that among 140 college students in South Korea, the average AI ethics awareness on a 5-point Likert scale was 3.39, which suggests that there is a moderate level of awareness. The results indicate that higher levels of digital literacy are linked with greater AI ethics awareness. Several studies indicate that awareness alone is not enough to combat the rise of sophisticated deepfakes; it is suggested to be accompanied by external support for its effectiveness. With that, it is essential to have structured interventions such as legislation, educational systems, and ethical and technical education (Tuysuz and Kilic, 2023; James et al., 2017; Ramachandran et al., 2024; Momeni, 2024). Ethical and technical education can increase individuals' moral sensitivity and empathy as it promotes responsible digital habits (Ramachandran, 2024). Deepfakes educational books include *Deepfakes: The Coming Infocalypse* by Nina Schick and *Deepfakes* by Graham Meikle. A simple game website called "Which Face is Real?" asks users which face is real from two pictures. The ethical position of an individual can reveal patterns in their moral reasoning and how this influences their decision-making and actions, notably in responding to deepfake content.

Detection Ability

Deepfakes are becoming more relevant as Artificial Intelligence technology advances, which makes it essential to understand the public's ability to detect them. Groh et al. (2021) conducted a study with over 15,000 respondents who guessed which images were fake and real. Out of the 242,216 guesses that were submitted, an average identification accuracy of 86% was achieved, with 88% of deepfakes correctly detected. It was discussed that detection accuracy is enhanced with exposure and experience. However, Rossler et al. (2019) indicate that detection accuracy is lower when studies use advanced synthetic face-swapping technologies. Participants scored 68.7% for high-quality images and around 58.7% for low-quality images. In Kobis and Soraperra's study, only 4 out of 16 deepfakes were identified correctly, with participants being substantially better at spotting authentic videos. This suggests a strong detection bias since they also guessed 'authentic' 67.4% of the time. These studies reveal that there is still a limitation in discernment between deepfakes and authentic media.

Detection of deepfakes becomes challenging when realistic fake content blends smoothly with real-world contexts to create deception, with the intent of producing fake news, making it difficult to identify which is true or false information (Egelhofer and Lechler, 2019). In line with this, a study by Hamellers et al. (2023) reveals that participants had a difficult time distinguishing between authentic and manipulated political speech. It was found that over 50% of participants doubted an authentic video, which indicates profound "false negatives". This is supported by the findings of iProov (2019) wherein people have struggled to identify fake content involving political figures, affecting the perception of authenticity and public trust.

As reported by the study of Alsobeh et al. (2024), there has been a significant correlation between cognitive biases, digital literacy, and the detection ability of deepfakes. It is argued in a similar study by Hameller et al. (2023) that pervasive overconfidence is a primary cognitive bias in detection ability. Overconfidence is one of the most costly biases and most common human decision-making errors (Kahneman, 2011). It heightens people's vulnerability to manipulation when detecting deepfakes. For instance, they might unknowingly consume fake content because they think it's genuine, influenced by their overconfidence. Overconfidence has been evident in the study of Kobis and Soraperra (2021) wherein, in most cases, their confidence is significantly higher than their actual accuracy. These findings imply that certain psychological factors can influence detection abilities.

Although a particular framework on deepfake detection is lacking, a relevant framework can be associated with it due to its alignment with the study's objectives. Signal Detection Theory (SDT) is a unified theoretical framework for studying fake news detection (Batailler et al., 2021). It has a long history of perceptual studies, especially in distinguishing signals from noise, especially in ambiguous situations (Abbas and Taeihagh, 2024). This is useful for understanding how individuals differentiate fake and real news in the context of deepfakes. This theory was applied to the study of Kobis and Soraperra (2021), wherein the participants had a modest accuracy of 59.6% for unbiased guesses on distinguishing real and fake videos.

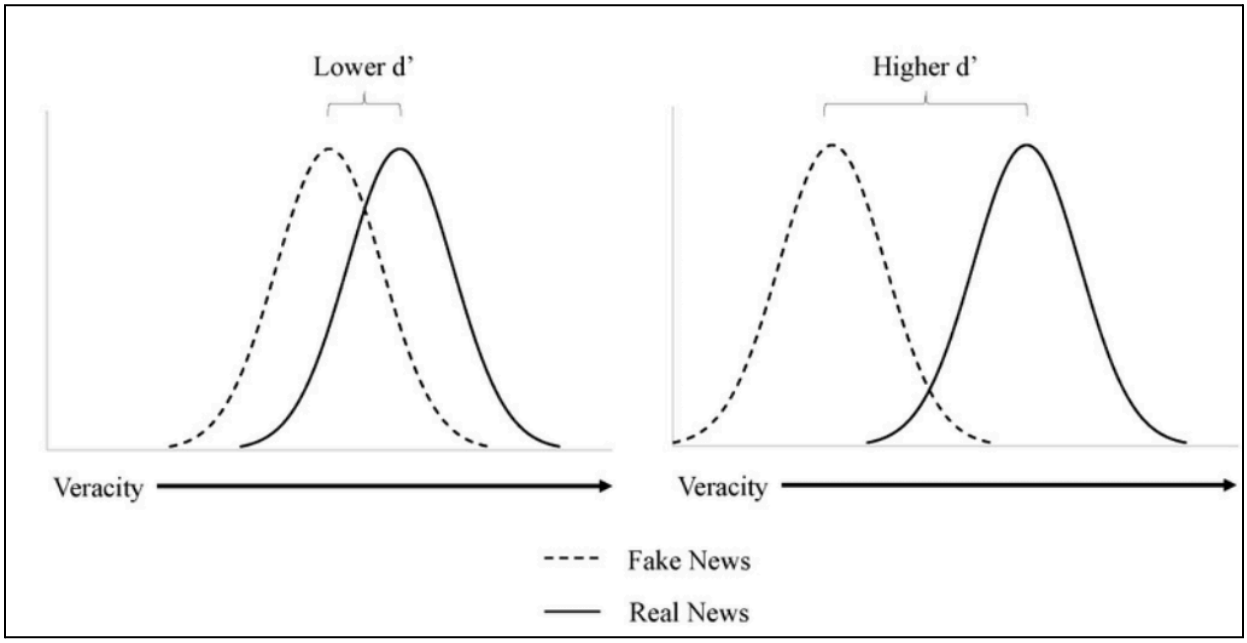


Figure 2.1. Signal Detection Theory's discrimination sensitivity (d') in distinguishing real from fake news

Figure 2.1 illustrates two distributions—real news and fake news with their veracity dimensions. When the distributions are far apart, people can easily differentiate real from fake news, leading to better detection accuracy and higher d' value or discrimination sensitivity. Conversely, if the distributions are closer or overlap, the detection accuracy and d' value are lower.

Reporting Behavior

Reporting malicious, unethical, and harmful content is a critical aspect of mitigating the impact of this emerging threat to maintain a safe online environment. The Theory of Planned Behavior (TPB) will serve as the foundational framework for understanding the users' intentions toward deepfakes on social media, which was also previously used in the study by Li and Zhao (2024).

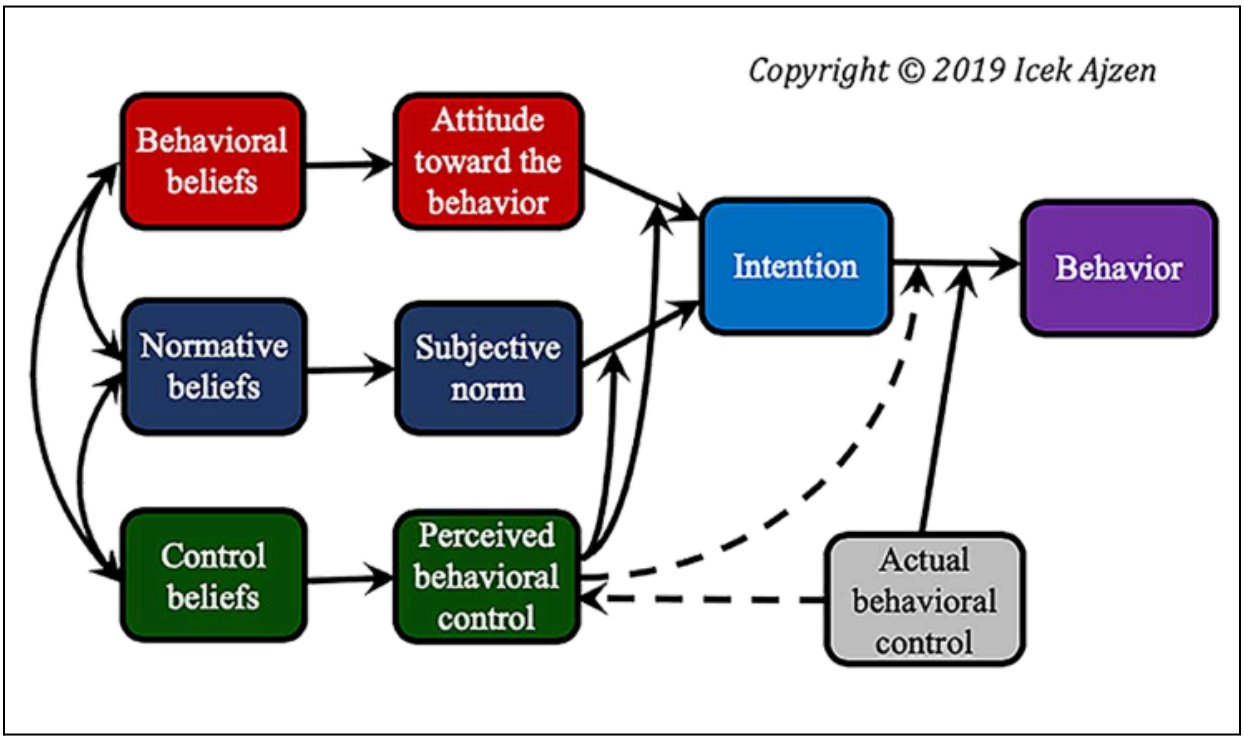


Figure 2.2. Schematic Representation of the Theory of Planned Behavior

The Theory of Planned Behavior states that an individual's behavior is driven by intention, which is shaped by their attitudes, subjective norms, and perceived behavioral control (Ajzen, 2019; Chen and Slade, 2024). These factors are essential to understanding individuals' willingness to report harmful deep fake content. According to the study by Wang et al. (2021), fake news posts with videos on social media platforms are more likely to be reported by individuals. Videos play an essential role in encouraging users to report the content and drive engagement, thereby amplifying the spread of misinformation. However, it was also discovered that fake news with strong

emotional intensity (positive or negative) and high popularity makes social media users less likely to report the content. Fake news content that triggers positive emotions or sentiments decreases the likelihood of individuals reporting the posts compared to negative ones. These findings indicate that the reporting behavior of users heavily depends on the sentiment or emotional intensity of the content, which opens vulnerability to exploitation by fake news creators. In a similar study by Vahed et al. (2024), it was explored how participants discern and respond to morally undesirable social media content. It was found that morally negative images have a higher likelihood of being reported by users (42.83%) than positive (0.26%) and neutral images (0.24%). Moral imagery is a strong factor that affects the participants' moral judgments. It was also highlighted that empathy positively influences reporting behavior. In other words, ethical awareness and empathy are key motivators for reporting harmful content like deepfakes.

The reporting mechanisms of social media platforms are important to know to evaluate their effectiveness in eradicating malicious content. An audit study by Qiwei et al. (2024) investigated the takedown speed for non-consensual intimate media on X (formerly Twitter). 50 AI nude images were generated, and half of them were reported under the Digital Millennium Copyright Act (DMCA) and half under Non-Consensual Nudity (NCN). It was revealed that within 25 hours, all 25 images under DMCA were removed promptly and received temporary bans and notifications for violation. Meanwhile, NCN reports under X's internal policy were not removed and remained visible during a three-week observation period. The account did not receive any notification or consequences of X. The findings highlight the noteworthy disparity in the effectiveness of reporting mechanisms or content removal on X. Not only on X, but social media platforms must have greater accountability and stronger legal mechanisms to ensure that malicious deep fake content is immediately removed and prevents its spread online.

Conceptual Framework

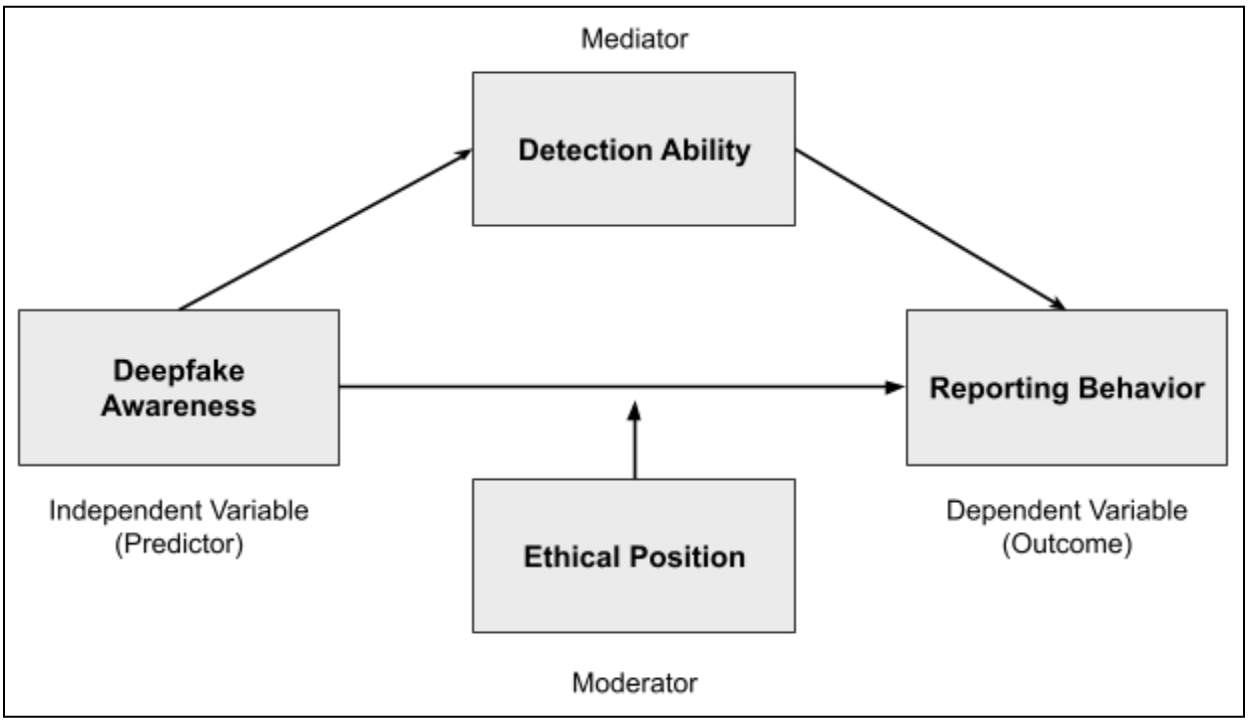


Figure 2.3. Conceptual Framework Portraying the Direct Influence of Deepfake Awareness on Reporting Behavior, with Detection Ability as a Mediator and Ethical Position as a Moderator.

The illustration shows a hypothetical relationship and path diagram between deepfake awareness, ethical position, detection ability, and reporting behavior. The arrows connecting different variables are presumed to influence one another, and each one of these effects, whether direct or indirect, will be tested in the study.

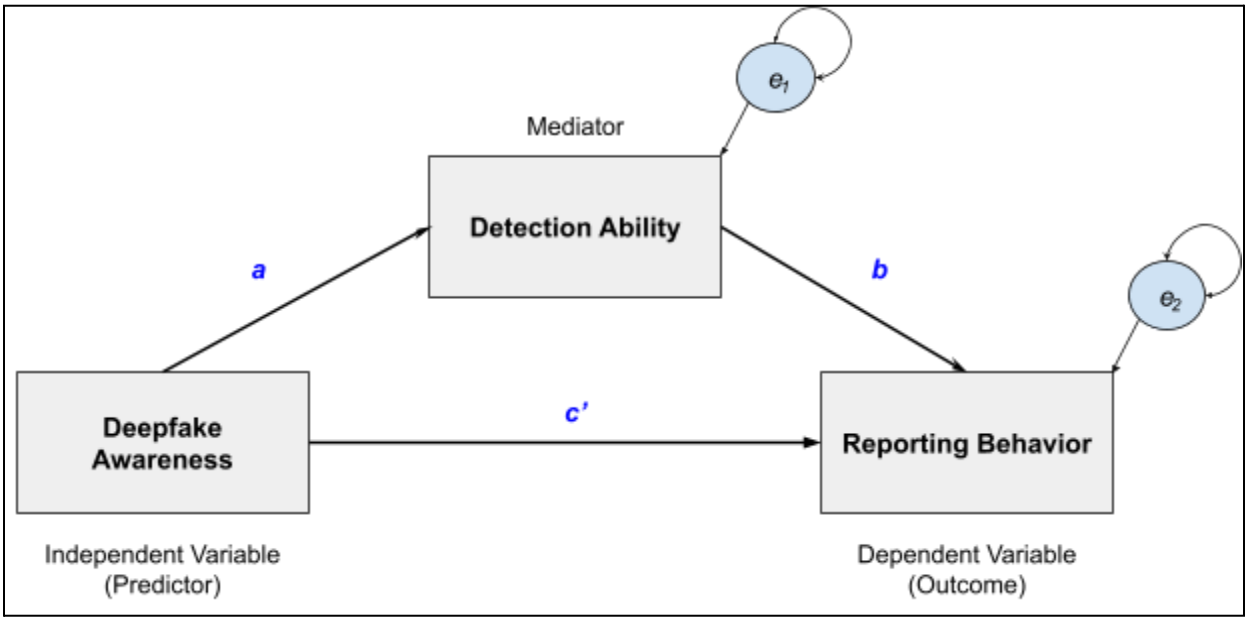


Figure 2.4. Mediation Model

The mediation model shows that deepfake awareness (predictor) directly influences reporting behavior (outcome), which represents path *c'* (direct effects). However, in some theories, awareness can indirectly affect behavior via the detection ability or the mediator, which represents the *a* and *b* pathways. The residual error is commonly depicted in the model through the notation of e_1 and e_2, which is crucial for regression analysis. A residual error is utilized to estimate how well the regression model fits the dataset (Bobbitt, 2021). The lower the residual error, the better a regression model fits a dataset. A mediator is a variable that explains the underlying mechanism that relates the two variables of interest, which would be tested using mediation analysis (Bhandari, 2021). The mediator is crucial for understanding the "how" or "why" behind the connection. Detection ability acts as the process variable that converts awareness into reporting action. For instance, a student equipped with both awareness and detection ability is more inclined to report malicious deepfakes. On the other hand, a person who has a high awareness of deepfakes but lacks detection ability may unknowingly scroll past a malicious, fake video and fail to report it.

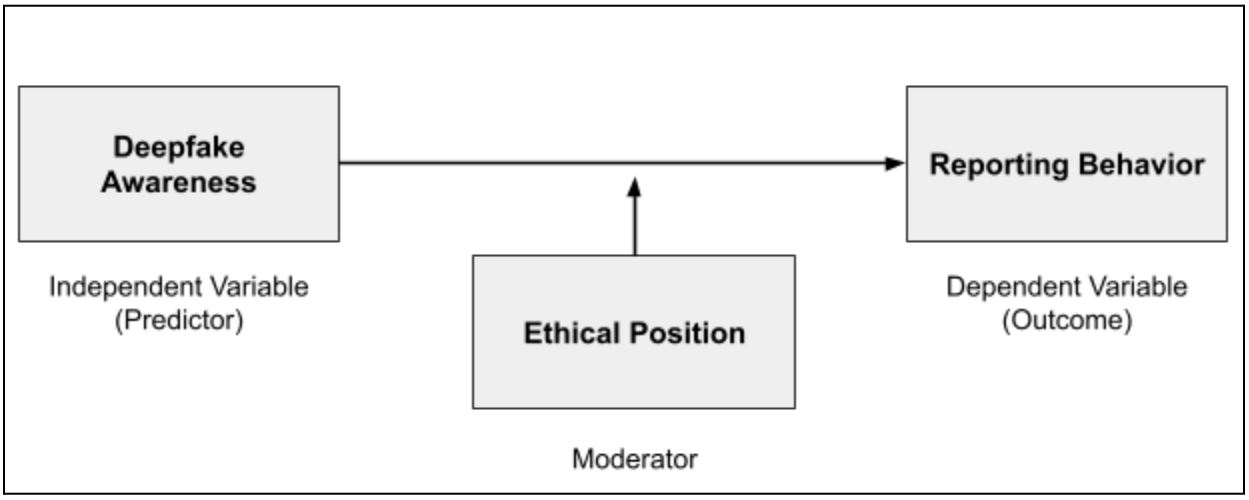


Figure 2.5. Moderation Model

Moreover, the study aims to explore the ethical view of the students by considering an ethical position as the moderator, wherein it influences the strength and direction of the relationship between deepfake awareness and reporting behavior. A moderator does not explain the underlying mechanism but instead how weak or strong the relationship between the two variables (McLeod, 2025). For example, the ethical position may suggest that a student who is absolutist (high idealism, low relativism) increases the likelihood of reporting deepfake content because they believe in obeying moral rules that produce a greater outcome for all. Despite high levels of deepfake awareness among individuals, their decision to report the malicious content may often be influenced by their ethical position. In short, the ethical position provides deeper insights to examine how varying moral perspectives may weaken or strengthen the connection between awareness and reporting behavior.

Methodology

Research Philosophy and Approach

A research philosophy and approach is a belief system or framework that guides the researcher on how they will conduct the study or investigation (Tamminen and Poucher, 2020). The study adopts positivism, which heavily relies on scientific methods and empirical evidence (Nickerson, 2023). Positivism emphasizes the core principle that only factual knowledge, mathematical proof, observations, and experiences are trustworthy, to ensure a logical and value-free approach (Philosophy Terms, 2016). However, positivism may overlook intricate details like time, space, and causation (Murzi, 2007). Since the study seeks to identify relationships among variables such as awareness, ethical position, detection, and reporting, a positivist framework aids the use of structured surveys and statistical analysis. This aligns with the principles of positivism—objectivity and empirical basis. Positivism is more suitable in this study compared to realism, interpretivism, and pragmatism because it highlights the use of questionnaires and statistical tools to test hypotheses on measurable data. However, it can ignore emotional and subjective elements, may oversimplify complex social phenomena, and lack understanding of lived experiences. There are several reasons why positivism is chosen. While realism also emphasizes objectivity, it supports both subjective interpretations and observable phenomena, which likely shifts the focus of the study. On the other hand, interpretivism leans more towards qualitative research that digs deeper into social constructs and personal meaning, making it less applicable in this study that is aiming for statistical generalization. Lastly, pragmatism supports practical outcomes and mixed methods, however, this research does not implement both qualitative and quantitative techniques, making it less aligned to pragmatism due to its purely quantitative structure.

Deductive reasoning is an applicable research approach for the study. Deductive reasoning utilizes top-down logic to make conclusions based on premises or hypotheses (Keating et al., 2019). It involves building on existing literature and theories. In this research, the approach is guided by existing theories, including the Theory of Planned Behavior and Signal Detection Theory, which provide the foundation for developing hypotheses about the relationships among the variables, such as mediation and moderation. Then, the hypotheses are tested through structured data collection and statistical analysis. Deductive reasoning was selected over abductive and inductive reasoning since it aligns with the positivist paradigm, supports a hypothesis-driven approach, tests pre-existing theories, and measures relationships using quantitative techniques. Yet, this approach may lead to confirmation bias or errors if assumptions are flawed and limited adaptability once hypotheses are set. While inductive reasoning also involves building theories from observations, it is more appropriate for exploratory research where the goal is to formulate new theoretical insights or where there is little existing literature. The researcher’s study is not exploratory, but rather seeks to validate

known frameworks in a specific context. On the other hand, abductive reasoning is frequently used when unexpected findings or complex phenomena arise, then researchers try to seek plausible explanations. This is common in qualitative and or mixed methods research, wherein it involves open-ended inquiry. This led the researcher to choose a deductive approach.

Methodological Choice and Strategy

In terms of methodology, the study adopts a quantitative research design. Quantitative research measures and analyzes relationships between variables, involving numerical data collection through structured surveys, and analyzes them statistically (Bhandari, 2020). This aligns with the positivist and deductive approaches. The study employs a correlational approach with causal inference techniques. While experimental or controlled interventions are absent, the researcher cannot establish definitive causality (DeCarlo et al., 2021). However, observational data from the study can provide insights into causal relationships when examined appropriately (Rohrer, 2018). This can be done through graphical causal models and by forming plausible assumptions supported by existing literature or theoretical frameworks. Quantitative research is chosen as well because it is suitable for testing linear regression, mediation and moderation using SPSS, increases the generalizability of findings among Metro Manila students, and allows standardized comparison between variables. A qualitative approach is ideal for exploring perceptions, personal experiences, and narratives through interviews or focus groups, however, it is not the goal of the study. It lacks the quantifiable metrics and structured measurement tools to establish relationships among variables and test hypotheses, which is the main strategy of the study. Since this research seeks to evaluate measurable constructs (awareness levels, ethical position, detection ability, likelihood to report), quantitative analysis is more appropriate, though quantitative methods may oversimplify human behaviour into numerical data and lack flexibility to modify during data collection.

Proceeding with the strategy, the sampling method will be non-probability sampling, which uses non-random criteria like geographical location, availability, or a particular characteristic of individuals (Kassiani, 2022). Within this category, the purposive sampling technique or snowball sampling is used. The researcher deliberately chooses or directly reaches out to respondents who are relevant to the research objective, specifically students currently enrolled in Metro Manila schools or universities, aged 18-34, and active social media users. This method is chosen for numerous reasons. It is very accessible given the researcher’s limited time frame and resources. It is also time-efficient, since it prevents the complexities of conducting large-scale outreach efforts or creating a random sample frame, which probability sampling would require. However, snowball sampling may introduce bias as it relies on personal networks, which could influence the diversity of the sample. This method may result in low generalizability to the wider population and may produce homogenous

samples, which reduces variability. On the other hand, probability sampling methods—simple random sampling, stratified sampling, and systematic sampling—require a complete and accurate sampling frame. This is impractical for the researcher's study, as it necessitates a list of all individuals in the target population. Such a list is not publicly available or easily accessible due to the numerous ethical or privacy concerns, data restrictions, and institutional boundaries. The researcher does not have institutional access to every school's student list, administrative permission to obtain them, nor the ability to choose respondents from the entire population randomly. This highlights the appropriateness of using non-probability purposive sampling rather than probability sampling.

Questionnaire

The questionnaires that will be used in the study are inspired by established theories and studies related to the variables. The researcher ensures that the instrument is grounded in academic research and peer-reviewed. Each question or item in the survey is designed to align with the study's variables, such as awareness, detection ability, reporting behavior, and some demographics. The researcher will combine the existing validated and reliable questionnaires and explore meaningful patterns of the data collected. The questionnaire that will be utilized in measuring and determining the ethical position of an individual is based on Forsyth's Ethics Position Theory (EPT). This theory assumes that individuals develop their unique intuitive moral beliefs, which focus on two main concerns: idealism and relativism (Forsyth, 2019). The theory originates from psychological studies on people's judgments of research ethics and aligns with philosophers' ethics based on deontology and consequentialism. Forsyth developed the Ethics Positions Questionnaire (EPQ), a 5-point Likert scale questionnaire, to evaluate personal moral philosophy where individuals fall on the scales of idealism and relativism to classify them into one of four ethical or moral positions: situationists, absolutists, subjectivists, and exceptionists (Forsyth, 2011).

	Low Relativism	High Relativism
High Idealism	**Absolutists:** Principled idealists who believe people should act in ways that are consistent with moral rules, for doing so will in most cases yield the best outcomes for all concerned	**Situationists:** Idealistic contextualists who favor securing the best possible consequences for all concerned even if doing so will violate traditional rules that define what is right and what is wrong
Low Idealism	**Exceptionists:** Principled pragmatists who endorse moral rules as guides for action, but admit that following rules will not necessarily generate the best consequences for all concerned	**Subjectivists:** Pragmatic relativists who base their ethical choices on personal considerations, such as individualized values, moral emotions, or an idiosyncratic moral philosophy

Source: Adapted from Forsyth, 1980

Figure 3.1. The Characteristics of the Four Moral Positions

The researcher will modify the questionnaire in the context of deepfakes, without losing its essence, in order to identify the ethical position of the student. The following scale is utilized to identify which category individuals fall into, based on the median scores from existing peer-reviewed studies: High Idealism (>3.7), Low Idealism (<3.7), High Relativism (>3.1), and Low Relativism (<3.1). Forsyth (1980) defines idealistic individuals as people who care for the well-being of others, while individuals who are inclined to relativism are those who do not base their values on universal ethical rules. Redesigning the questionnaire in the context of deepfakes will help the researcher provide insights into the moral position of the students.

On the other hand, a survey created by Bitton et al. (2021) on the researcher's study, "*Deepfakes in the context of AI inequalities: analysing disparities in knowledge and attitudes*", will be utilized to measure deepfake awareness among students. The questionnaire includes sections in the form of a Likert Scale about deepfake awareness and knowledge, perceptions of risks and opportunities, and digital skills and innovativeness. The researcher will only ask questions that are relevant to the study's objectives and variables. The survey was conducted with 1,421 German internet users, and it was revealed that there was generally a low level of deepfake knowledge, and this knowledge affects their positive attitudes towards the technology. In terms of measuring the detection ability, a questionnaire developed and validated by Plohl et al. (2024) will be used, namely, the Perceived Deepfake Trustworthiness Questionnaire (PDTQ). The study focuses on developing a questionnaire that allows the researchers to assess an individual's perceived trustworthiness of deepfake videos on a Likert Scale. Some of the constructs that are being measured in the questionnaire are persuasiveness, vividness, credibility, authenticity of the message, and perceived video fakeness. The researcher will present a mix of real and fake video samples and have the respondents rate each of their authenticity. Lastly, the reporting behavior of an individual will be measured by constructing a questionnaire based on the Theory of Planned Behavior, which was defined in the literature review. Ajzen (2019) constructed a comprehensive format of a Likert scale questionnaire based on the study of Fishbein and Ajzen (2010), which discusses the Theory of Reasoned Action approach, or the older version of the Theory of Planned Behavior. In the context of this study, the researcher will specifically design the questionnaire on reporting behavior toward deepfakes in order to understand how different factors interact to influence the actions or intentions of a student. The questionnaire assesses behavioral beliefs, attitudes, normative influences, behavioral control, and intentions.

Data Collection

The researcher will utilize survey research as a systematic data collection method to collect quantitative data from a sample (Hassan, 2024). This involves gathering information about individuals' behaviors, experiences, demographics, and decisions. The type of survey research combines analytical and cross-sectional

approaches, which will be created, distributed online, and self-administered. The mode of administration will involve the dissemination of the Google Forms survey through communication channels and social media platforms, making it efficient and accessible for the target sample. This eliminates the costs for printing and data entry. Interviews/focus groups are not conducted since it is time-consuming, lack statistical depth, and are less scalable. Experiments are not considered since they possibly require higher costs, complexity, and stringent ethical considerations. This is why the survey is one of the best data collection methods due to its cost-effectiveness, comprehensiveness, and flexibility (Balamurugan, 2023). It is easily quantifiable and analyzable. However, surveys may not entirely capture the depth of personal motivations or the depth of ethical reasoning behind reporting behavior. Since detection ability is measured through visuals, it can present some limitations in terms of device quality or screen size. Survey may have self-report bias and potential sampling bias due to the non-probability approach. Common pitfalls include the risk of non-serious responses, particularly in detection tasks, technical difficulties, and a lack of control over the environment where respondents accomplish the survey.

The questionnaire will involve a few demographic questions to confirm their eligibility for the study, add context to the data analysis, and provide a brief background about the respondents. A consent form will be attached to the first page of the questionnaire (Appendix B) to inform the respondents of their rights, consent, and confidentiality. The researcher ensures anonymity and confidentiality of the respondents' data throughout the research process to ensure ethical treatment. Participation was voluntary, and the questionnaire was designed to prevent biased or distressing questions. The minimum required sample size is 55 in order to conduct multiple regression, according to the sample size calculator (Soper, 2025).

Pilot Testing and Reliability Test

Prior to final data collection, a pilot test involving 10 respondents will be conducted to evaluate the internal consistency of the entire survey using the Cronbach Alpha reliability test through SPSS. Moreover, the pilot test aims to collect open-ended feedback for clarity, such as determining confusing questions or sections of the survey (e.g., "Were there any sections of the survey that are confusing?") (Appendix B). According to Tappin (2014), historical studies recommend that the minimum number of respondents for pilot testing is from 10 to 30. The following steps were undertaken for the pilot testing:

1. The researcher communicated with 10 eligible respondents via Facebook and invited them to participate in the pilot testing.
2. The survey was distributed online for ease of access and included a section clearly outlining consent, data protection, an overview, and guidelines.
3. Open-ended feedback results revealed that:

- 9 out of 10 respondents indicated that none of the survey questions were confusing.
- 1 respondent mentioned that some questions in the detection ability sections were inconsistent
- 2 respondents commented that the survey was too lengthy.

Overall, the majority of the respondents agreed that the survey was neither too long nor too short, and no technical issues were reported. After gathering the responses, Microsoft Excel was used to preprocess the data by translating Likert scale test results into numeric values (e.g., “Frequently” was coded as “5”) to ensure uniformity and preparation for reliability analysis. The cleaned data were then imported into SPSS software, where appropriate data types were assigned. The Cronbach Alpha reliability test was conducted on four scales. According to Fein (2022), this analysis indicates that removing particular items from the questionnaires improves the reliability score of the scale. According to Frost (2022), a rule of thumb for interpreting Likert scale or dichotomous questions is as follows:

Cronbach's alpha	Internal consistency
$\alpha \geq 0.9$	Excellent
$0.9 > \alpha \geq 0.8$	Good
$0.8 > \alpha \geq 0.7$	Acceptable
$0.7 > \alpha \geq 0.6$	Questionable
$0.6 > \alpha \geq 0.5$	Poor
$0.5 > \alpha$	Unacceptable

Figure 3.2. *Cronbach Alpha Scores’ Levels*

The resulting Cronbach Alpha reliability scores for each questionnaire are as follows (see Appendix D for detailed reliability results):

1. Scale 1 *(Deepfake Risk and Opportunity Perception Questionnaire)*: Initially scored 0.803 (20 items), showing acceptable reliability. Removing 2 items boosted the score to 0.855 (18 items in total), resulting in higher reliability.
2. Scale 2 *(The Forsyth’s Ethics Position Questionnaire)*: Originally scored 0.517 (13 items), indicating poor reliability. Excluding 5 items significantly improved the reliability to 0.832 (8 items in total), which is considered acceptable.
3. Scale 3 *(Perceived Deepfake Trustworthiness Questionnaire)*: Initially scored 0.800 (12 items), which demonstrates acceptable reliability. After removing 3 items, the score improved to 0.941 (9 items in total), which falls within the excellent range.

4. Scale 4 *(The Theory of Planned Behavior Questionnaire)*: Started with a very low score of 0.208 (11 items), reflecting unacceptable reliability. Excluding 6 items, the score greatly increased to 0.812 (5 items in total), achieving an acceptable level.
5. Moreover, the researcher tested the reliability of the entire survey or the combined scales. The resulting Cronbach Alpha score was 0.883, which indicates good internal consistency. Based on these reliability results, the survey is deemed ready for final data collection.

Analytical Techniques

The analytical techniques to be used are Mediation analysis, Moderation analysis, and Linear Regression, with the use of statistical software called IBM SPSS. The researcher initially considered Structured Equation Modeling (SEM) to thoroughly investigate causal relationships among variables, however, it is not feasible due to the assumption that it requires at least a sample size of 200 for accurate results (Akinremi, 2024). IBM SPSS or Statistical Package for the Social Sciences is a paid software that is commonly used for research and advanced statistical analysis, including frequencies, bivariate statistics, cross-tabulation, and linear regression (Williams, 2025). The Hayes PROCESS Macro is a freely-available computational tool for SPSS that conducts the mediation and moderation analyses (SPSS, 2024). Though SPSS is paid and licensed software, the researcher subscribed to 1-month free access to reduce the direct cost. SPSS is chosen because it is user-friendly for statistical testing and is robust for mediation and moderation analysis. However, some assumptions (e.g., linearity, normality) must be satisfied for reliable and accurate interpretation. Common pitfalls include excessive reliance on p-values and misinterpretation of statistical results without a solid background. On the other hand, Python was used as a complementary tool to perform descriptive statistics and conduct exploratory data analysis. This is to compute for the summary statistics (e.g., means, frequencies, and standard deviations), and to visualize respondent distributions and response trends for each variable. Using Python enables the researcher for preprocessing and filtering, which assists in verifying data integrity and determining preliminary patterns.

Mediation analysis quantifies the causal sequence by understanding how one variable influences another through a mediator variable (Oxford Bibliographies, 2019). It helps determine whether a mediator exists in the relationship. In this study, mediation analysis checks how deepfake awareness affects detection ability and how detection ability, in turn, affects an individual's reporting behaviour toward deepfake content, which was explained in the conceptual framework. The study examines how much the effect of awareness on reporting behaviour is mediated through detection ability. Mediation analysis does not represent causality, but it is a tool to explore causal relationships. In this study, the type of mediation that will be used is partial mediation, wherein the direct relationship (c') between independent and dependent variables in the

presence of a mediator is estimated to be a non-zero coefficient (Jung, 2021). To test for the presence of mediation, the researcher will estimate the effect of the indirect effect and whether it has significance in the relationship.

To examine the influence of the moderator variable on the predictor and outcome variables, it is essential to apply moderation analysis (also known as effect modification). One of the key steps in testing moderation is to run a multiple regression model in order to predict the dependent variable and examine the plot to see if the relationship between the two variables changes at different levels of the moderator. This is to check if the effect of deepfake awareness on reporting behavior is different in varying ethical positions. If the interaction shows significance, it means that the relationship between the predictor and outcome may depend on the moderator. However, this doesn't imply causal sequence, unlike mediation analysis.

Given all of these analytical techniques, it is important to determine how the researcher will conduct these. The researcher will conduct regression analysis, since it analyzes and models the relationship between the dependent and independent variables (GeeksforGeeks, 2017). The goal of this method is to identify the most suitable function that distinguishes the connection between these variables, which draws conclusions or enables predictions. In the study, it is appropriate to run regression models to evaluate the effect of path *a*, path *b*, path *c*', and *a x b* or indirect effect. Incorporating linear regression will examine the relationship between deepfake awareness (predictor) and reporting behavior (outcome), or the direct effects. Linear regression, as a foundational statistical technique, is central to this study as it models the relationship between the key variables: independent and dependent. Furthermore, regression analysis confirms to be particularly useful for conducting mediation and moderation analyses.

Timeline

The research will span seven months, starting from November 2024 until May 2025, with specific objectives for each stage. From November to January 2025, the focus will primarily be on the topic proposal, including the background of the research, a review of related literature, and the methodology. From February to April 2025, data gathering, analysis, and interpretation of results will be done. Lastly, May 2025 is the completion of the research report and presentation of findings.

Results and Findings

A total of 60 survey responses were collected from students across Metro Manila. According to Soper (2025), the minimum required sample size is 55 in order to conduct multiple regression, according to the sample size calculator. Thus, the collected data is more than sufficient. The dataset was analyzed using SPSS and Python in order to examine the relationships between deepfake awareness, detection ability, ethical position, and reporting behavior. Analytical techniques such as descriptive statistics, linear regression, mediation, and moderation analysis will be covered in this section.

Descriptive Statistics

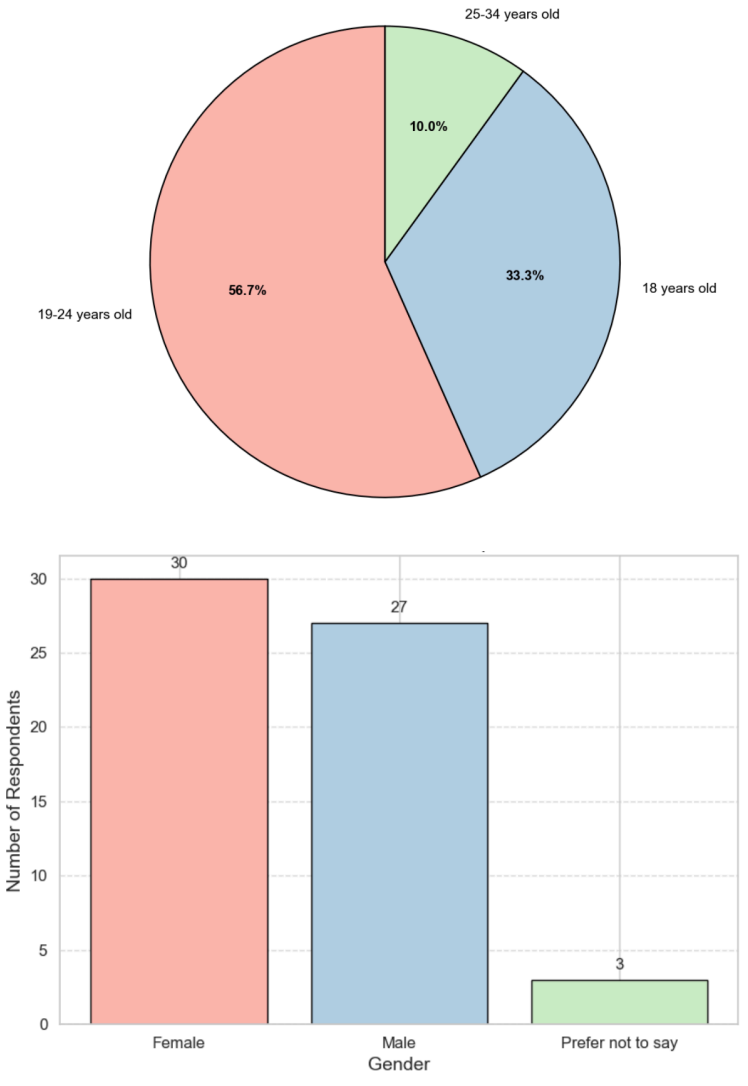


Figure 4.1. Age Group and Gender Distribution of Respondents

65% of the respondents were aged 19–24 years old (n = 39), followed by 18-year-olds (28.3%, n = 17), and a small portion (6.7%, n = 4) were between the ages of 25–34 years old. The gender distribution is fairly balanced, with 55% identifying as female (n = 33) and 40% as male (n = 24), while 5% of the respondents preferred not to disclose (n = 3). Most of the respondents are college/undergraduate students (n = 35, or 58.3%).

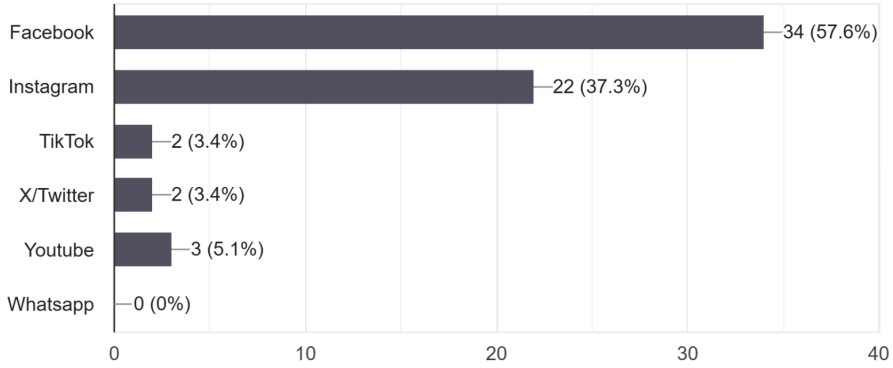


Figure 4.2. Most Used Social Media Platforms

The majority of the respondents were highly active on Facebook (57.6%, n = 34) and Instagram (37.3%, n = 22). This is followed by YouTube (5.1%, n = 3), TikTok and X (3.4%, n = 2), and none are WhatsApp users. In terms of the frequency of usage of social media, 50% indicated the highest level (rating 5), and 30% gave a rating of 4, which means that most of them are daily social media users.

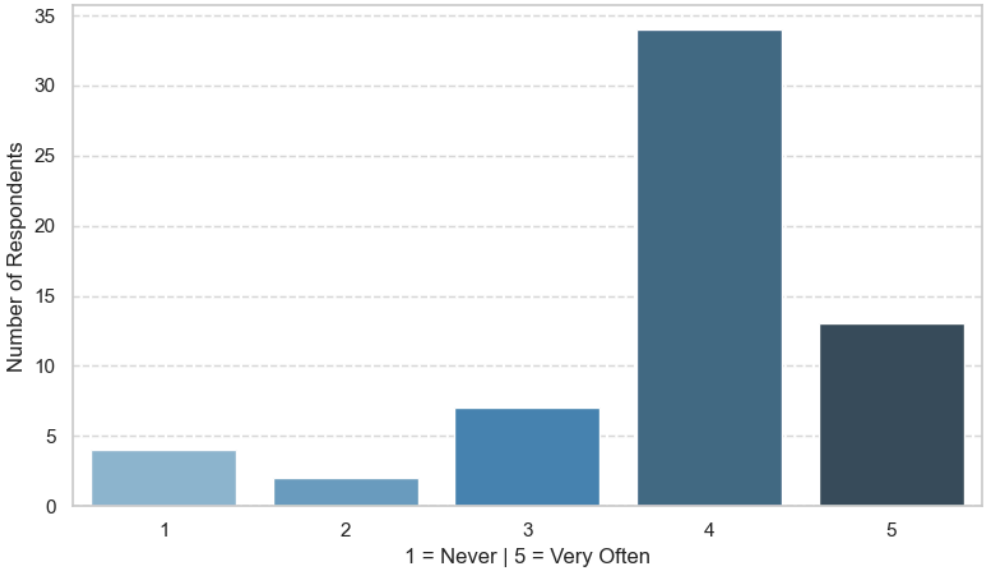


Figure 4.3. Frequency of Using AI Tool Usage

When asked about comfort on digital tools, 43.3% (n = 26) of the respondents rated themselves at 4, and 20% (n = 12) rated 5. This comfort extended to AI tools (e.g., ChatGPT, voice assistants, and image generators), with 78.4% (n = 47) of the respondents using them frequently.

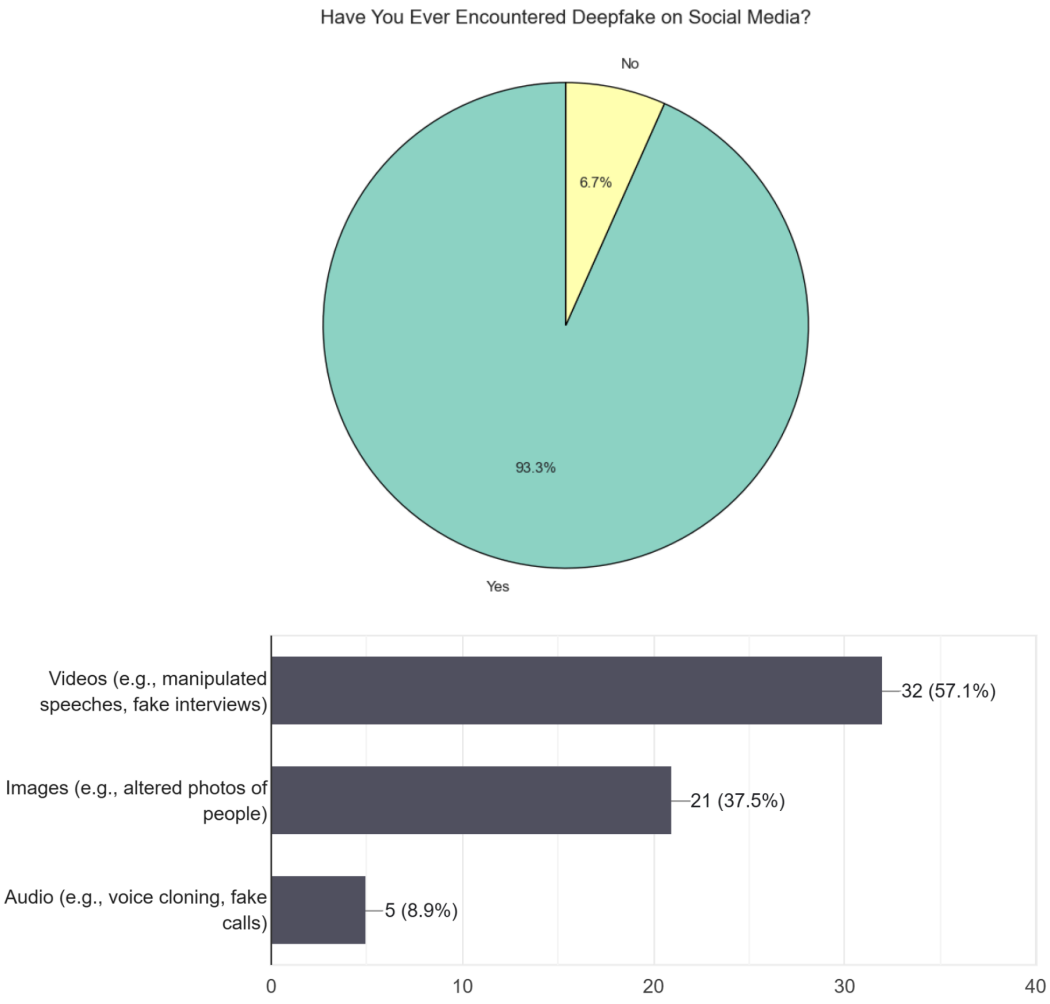


Figure 4.4. Deepfake Exposure of Respondents

A notable number of respondents (93.3%, n = 56) reported that they have encountered deepfakes on social media. Among those, 57.1% watched deepfake videos (e.g., manipulated speeches, fake interviews), 37.5% encountered altered images or photos, and 8.9% heard voice-based deepfakes (e.g., voice cloning, fake calls). Regarding frequency, 81.7% (n = 49) rated their exposure to fake or manipulated content at 4 or 5, which indicates they regularly encounter deepfakes online.

	count	mean	std	min	25%	50%	75%	max
concern_falling	60.0	2.750000	0.600141	1.0	3.0	3.0	3.00	3.0
concern_victim	60.0	2.700000	0.618911	1.0	3.0	3.0	3.00	3.0
concern_politics	60.0	2.533333	0.623459	1.0	2.0	3.0	3.00	3.0
concern_trust	60.0	2.283333	0.903696	1.0	1.0	3.0	3.00	3.0
concern_news	60.0	2.650000	0.633125	1.0	2.0	3.0	3.00	3.0
concern_easy_creation	60.0	2.500000	0.596601	1.0	2.0	3.0	3.00	3.0
concern_data_manip	60.0	2.633333	0.609714	1.0	2.0	3.0	3.00	3.0
concern_data_protect	60.0	2.516667	0.596364	1.0	2.0	3.0	3.00	3.0
benefits_general	60.0	2.383333	0.715250	1.0	2.0	3.0	3.00	3.0
benefits_fun	60.0	1.966667	0.609714	1.0	2.0	2.0	2.00	3.0
benefits_accessible	60.0	1.816667	0.812856	1.0	1.0	2.0	2.25	3.0
benefits_film	60.0	1.716667	0.454420	1.0	1.0	2.0	2.00	2.0
benefits_fashion	60.0	1.600000	0.588016	1.0	1.0	2.0	2.00	3.0
benefits_creativity	60.0	1.666667	0.705106	1.0	1.0	2.0	2.00	3.0
benefits_socialmedia	60.0	1.716667	0.761169	1.0	1.0	2.0	2.00	3.0

Table 4.1. Summary Statistics of Deepfake Awareness

The deepfake awareness is measured through respondents' concerns and perceived benefits about deepfake technology through a 3-point Likert scale (1 = Completely or rather disagree, 2 = Neither agree nor disagree, 3 = Completely or rather agree). The majority of the respondents expressed high concerns or strongly agreed with the risk of falling for a deepfake (M = 2.75, SD = 0.60). They also showed a high level of agreement with the concern about becoming a victim of deepfakes, such as scams and voice cloning (M = 2.70, SD = 0.62). Aside from that, they also worried about the deepfake misuse on political aspects (M = 2.53, SD = 0.62), data manipulation (M = 2.63, SD = 0.61), and the spread of manipulated news (M = 2.65, SD = 0.63). The lowest concern among all the items was the possibility of loss of trust in video form content due to deepfakes (M = 2.28, SD = 0.90), which also has the highest standard deviation, indicating mixed perspectives. In contrast, the mean scores for the perceived benefits of deepfake technology are lower than the concerns, which suggests that respondents have scepticism or a lack of perceived value in the positive usage of deepfake technology. The highest scored benefit was the idea that deepfakes can provide general benefits (M = 2.38, SD = 0.72). Respondents were relatively neutral in terms of entertainment (M = 1.97, SD = 0.61) value and ease of access (M = 1.82, SD = 0.81). The lowest scored benefits were connected to the application of deepfakes to specific fields, including fashion (M = 1.60, SD = 0.59) and film (M = 1.72, SD = 0.75).

	count	mean	std	min	25%	50%	75%	max
detection_accuracy	60.0	0.746667	0.235398	0.000000	0.6	0.800000	1.000000	1.000000
confidence_score	60.0	3.396667	0.543051	2.000000	3.0	3.200000	3.700000	4.600000
perceptual_score	60.0	3.705238	0.620674	2.514286	3.4	3.828571	4.057143	5.514286

Table 4.2. Summary Statistics for Detection Accuracy, Confidence Score, and Perceptual Score

In order to assess the detection ability, three components were used: detection accuracy, confidence score, and perceptual score. The detection accuracy is the proportion of correctly classified videos as real or fake out of 5 total videos. To process this data, it converts “Real” and “Fake” responses into numerical values, 1 and 0, respectively. Then, it is compared with predefined correct answers to compute detection accuracy for each video and the total average accuracy per participant. The confidence score is to measure how confident the respondents were in their judgements (1 = Not confident at all, 5 = Very confident). Finally, the perceptual score is a composite score that reflects their awareness or sensitivity to detection cues (e.g., lip-sync issues, facial distortions, strange mouth movements), based on 7-point Likert scale items per video. Starting with detection accuracy, respondents showed relatively high detection accuracy in identifying real and fake videos (M = 0.75, SD = 0.24). On average, respondents correctly identified 3.75 out of 5 videos. On the other hand, respondents demonstrated moderate to high levels of confidence in their judgements (M = 3.40, SD = 0.54), wherein scores ranged from 2.0 to 4.6, with a median confidence at 3.2. Among the three detection indicators, the perceptual score had the highest mean (M = 3.71, SD = 0.62). This suggests that respondents moderately recognized signs of deepfakes, despite their final classification being incorrect.

Video	True Label	Correct Responses	Incorrect Responses	% Correct	Average Confidence
Video 1	Real	40	20	66.7%	3.47
Video 2	Fake	47	13	78.3%	3.32
Video 3	Fake	37	23	61.7%	3.18
Video 4	Real	53	7	88.3%	3.55
Video 5	Fake	47	13	78.3%	3.47

Table 4.3. Detection Accuracy and Confidence Levels for Real and Fake Videos

The table presents the breakdown of correct and incorrect classifications of videos, along with the average confidence of the respondents. Video 4 (Real) was the most accurately identified video with 88.3% accuracy and the highest confidence level of 3.55. Meanwhile, Video 3 (Fake) has the lowest detection accuracy (61.7%) and the lowest confidence level (3.18), which suggests it was less obviously fake and more deceptive than the rest of the videos.

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
ethics_dignity	60.0	NaN	NaN	NaN	3.933333	1.117908	1.0	3.0	4.0	5.0	5.0
ethics_varies	60.0	NaN	NaN	NaN	3.25	0.950022	1.0	3.0	3.0	4.0	5.0
ethics_individual	60.0	NaN	NaN	NaN	3.133333	1.016252	1.0	2.0	3.0	4.0	5.0
ethics_unsolvable	60.0	NaN	NaN	NaN	3.066667	1.006195	1.0	3.0	3.0	4.0	5.0
ethics_personal_rules	60.0	NaN	NaN	NaN	3.466667	0.98233	1.0	3.0	4.0	4.0	5.0
ethics_own_code	60.0	NaN	NaN	NaN	2.983333	0.892372	1.0	2.0	3.0	4.0	5.0
ethics_political_satire	60.0	NaN	NaN	NaN	2.533333	1.47828	1.0	1.0	2.0	4.0	5.0
ethics_entertainment	60.0	NaN	NaN	NaN	2.2	0.970759	1.0	1.0	2.0	3.0	5.0
ethics_distress	60.0	NaN	NaN	NaN	2.433333	1.280448	1.0	1.0	2.0	3.0	5.0
ethics_choice1	60	2	Sometimes, even the best use of deepfake techn...	31	NaN	NaN	NaN	NaN	NaN	NaN	NaN
ethics_choice2	60	2	Moral standards regarding the use of deepfakes...	36	NaN	NaN	NaN	NaN	NaN	NaN	NaN

	count	mean	std	min	25%	50%	75%	max
idealism_score	60.0	2.775	0.825571	1.75	2.0	2.375	3.75	4.25
relativism_score	60.0	3.180	0.417742	1.80	3.0	3.200	3.45	3.80

Table 4.4. Summary Statistics of Ethical Position

The majority of the respondents agreed that deepfakes must respect human dignity (M = 3.93, SD = 1.12). There is a moderate agreement between respondents that individuals should follow personal moral rules in judging deepfakes (M = 3.47, SD = 0.98). It also shows a low support or disagreement with justifying deepfakes for entertainment (M = 2.20, SD = 0.97) or political satire (M = 2.53, SD = 1.48). Respondents demonstrate ethical concerns about deepfakes in terms of psychological harm (M = 2.43, SD = 1.28). The ethical ideologies of the participants were determined by measuring through two primary dimensions: Idealism and Relativism. The scores were used to segment participants into one of four ethical positions: Absolutist, Situationist, Exceptionist, or Subjectivist. Respondents exhibited moderate levels of idealism (M = 2.78, SD = 0.83), while they tended to lean more towards high relativism (M = 3.18, SD = 0.42).

The median-split method is a way to segment respondents into “high” and “low” groups based on whether their scores are categorized above or below the median of a particular variable. First, the average or total scores for Idealism and Relativism per respondent were computed. Then, scores of each of the respondents were compared against the following median thresholds or cutoff points (Forsyth et al., 2008). With the use of the median-split method, participants were grouped into the following ethical positions.

Ethical Position	Idealism	Relativism	Description	Number of Respondents
Absolutist	High (> 3.7 / 18.5)	Low (≤ 3.1 / 15.5)	Strong belief in moral absolutes and ideal outcomes.	6
Situationist	High (> 3.7 / 18.5)	High (> 3.1 / 15.5)	Ethics depend on the situation, but aim for ideal outcomes.	13
Exceptionist	Low (≤ 3.7 / 18.5)	Low (≤ 3.1 / 15.5)	Moral rules are guidelines, not absolute.	15
Subjectivist	Low (≤ 3.7 / 18.5)	High (> 3.1 / 15.5)	Moral decisions are based on personal feelings.	26
		Total		60

Table 4.5. Classification of Ethical Positions Based on Median-Split of Idealism and Relativism Scores

It is evident that the most common ethical position was Subjectivism (n = 26), followed by Exceptionism (n = 15) and Situationism (n = 13). Meanwhile, only 6 respondents were classified under Absolutism, which indicates that a very small number of respondents support strong moral rules and are highly concerned for ideal outcomes.

	count	mean	std	min	25%	50%	75%	max
report_helpful	60.0	5.233333	1.608944	1.0	4.0	6.0	6.25	7.0
report_importance	60.0	5.033333	1.506671	1.0	5.0	5.0	6.00	7.0
report_social_pressure	60.0	5.200000	1.259540	1.0	5.0	5.0	6.00	7.0
report_intention	60.0	5.400000	1.416609	1.0	5.0	6.0	6.00	7.0
report_celebrity	60.0	4.800000	1.527155	2.0	3.0	5.0	6.00	7.0

Table 4.6. Summary Statistics of Reporting Behavior

75% of the respondents strongly believe that reporting helps reduce misinformation (M = 5.23, SD = 1.61), and they generally agreed that the act of reporting is important (M = 5.03, SD = 1.51). Respondents also feel social pressure or are expected to report deepfake videos (M = 5.20, SD = 1.26). Most importantly, with the highest mean of 5.40 (SD = 1.42), they expressed a very strong personal commitment to report upon encountering deepfakes. Although the overall willingness is relatively high, some respondents may be less inclined to report when the particular deepfake targets a celebrity. This reflected in slightly higher variability, with a mean of 4.80 (SD = 1.53).

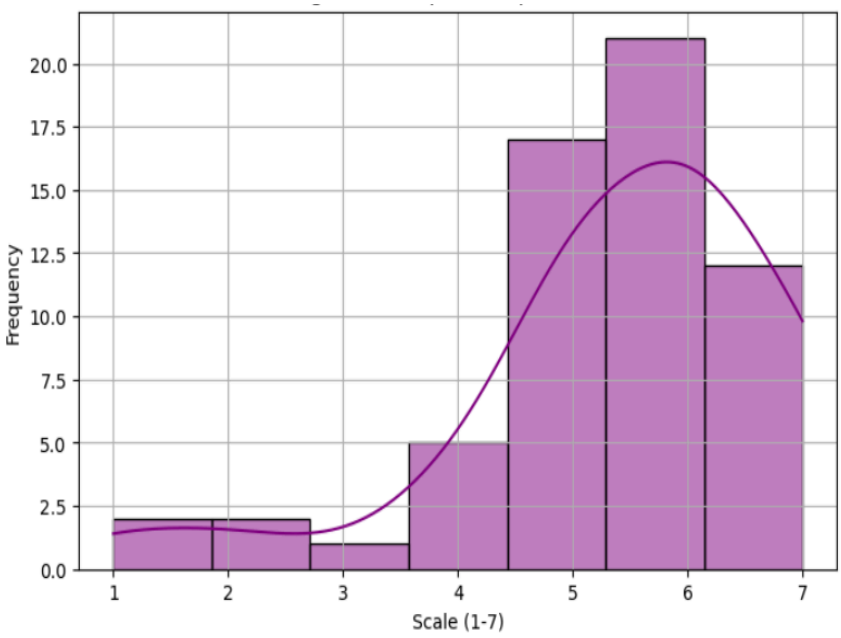


Figure 4.5. Histogram of Willingness to Report Deepfake Content

It reveals that there is a positive disposition among respondents toward reporting deepfake content. The distribution of the histogram is right-skewed, which indicates that the majority of respondents are clustered toward the higher end of the 7-point Likert scale (values 5 to 7), with very few selecting the lower values (1 to 3). This suggests the strong willingness of respondents to report deepfake content.

Inferential Statistics

Simple Linear Regression Analysis

A simple linear regression was conducted using SPSS to identify whether deepfake awareness significantly predicts reporting behaviour. The predictor variable was the computed Awareness_Score, obtained by averaging all responses to 15

deepfake awareness Likert scale items. On the other hand, the outcome variable was the Reporting_Score, computed by averaging responses to 5 reporting behaviour Likert scale items.

Table 4.7. Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.511[a]	.262	.249	.79510

a. Predictors: (Constant), Awareness_Score
b. Dependent Variable: Reporting_Score

The regression model was found to be statistically significant. With an R of .511, it suggests that there is a moderate positive correlation between deepfake awareness and reporting behaviour. The coefficient determination (R^2 = .262) means that approximately 26.2% of the variance in reporting behaviour can be explained by deepfake awareness alone. The adjusted R^2 explains for possible overfitting and offers a better estimate of the accuracy of the model's explanatory power for the general population. Based on the findings, it is reasonable to state that, although awareness significantly predicts reporting behaviour, other unidentified factors may also contribute to the remaining 73.8% of the variance.

Table 4.8. ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	12.986	1	12.986	20.542	<.001[b]
	Residual	36.667	58	.632		
	Total	49.653	59			

a. Dependent Variable: Reporting_Score
b. Predictors: (Constant), Awareness_Score

The ANOVA results show that the entire regression model is statistically significant, $F(1,58) = 20.542$, $p < .001$. This verifies that the relationship between deepfake awareness and reporting behaviour is unlikely to be random or due to chance, and that the independent variable notably improves the model's predictive power for the dependent variable.

Table 4.9. Regression Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			95.0% Confidence Interval for B	
		B	Std. Error	Beta	t	Sig.	Lower Bound	Upper Bound
1	(Constant)	1.362	.838		1.625	.110	-.316	3.040
	Awareness_Score	1.692	.373	.511	4.532	<.001	.945	2.439

a. Dependent Variable: Reporting_Score

Based on the unstandardized regression coefficient (B = 1.692), it implies that for each 1-point increase in deepfake awareness, the predicted reporting behaviour score rises by around 1.692 points, with the assumption that the other variables remain constant. On the other hand, the standardized coefficient (β = .511) reflects a moderate effect size, which shows that awareness is a fairly strong predictor of the intention to report deepfake content. The p-value < .001 states that the effect of deepfake awareness on reporting behaviour is statistically significant at the 0.5 level, which confirms this relationship is unlikely to be due to random variation. Moreover, the 95% confidence interval for B ranges from 0.945 to 2.439, emphasizing the precision and reliability of the estimate. To sum up, the findings suggest a statistically significant and moderately strong predictive relationship between deepfake awareness and reporting behaviour. This can imply that respondents who exhibited higher levels of awareness of deepfake technology were more likely to express stronger intentions to report such content online.

Mediation Analysis

A mediation analysis was performed using PROCESS Macro Version 5.0 by CCRAM expert, Hayes (2023), to test whether detection ability (M) mediates the relationship between deepfake awareness (X) and reporting behaviour (Y). Model Number 4 was selected to create a simple mediation model. The number of bootstrap samples is 5000, and the confidence interval is 95%. This model shows the total effect, direct effect, and indirect effects (with bootstrapping confidence intervals) and significant tests (p-values) for each pathway. To compute the final detection ability score (Detection_Score), 5 new variables were created and averaged (Accuracy1 to Accuracy5), which check whether the participants classified each video correctly. The Detection_Score will serve as the mediator for the analysis.

Table 4.10. Effect of Awareness on Detection Ability (Path *a*)

Predictor	B	SE	t	p	95% Confidence Interval	
					Lower Limit	Upper Limit
Constant	.256	.242	1.058	.294	-.228	.740
Awareness	.220	.108	2.046	.045	0.005	.436

In the first regression model, the effect of deepfake awareness on detection ability was tested. The table showed a significant positive effect of awareness on detection ability (B = .220, SE = .108, t(58) = 2.046, p =.045). It confirms that the effect is statistically significant, since the 95% confidence interval does not include zero. To state, every one-unit increase in awareness corresponds to a .220 increase in detection ability. This implies that students with greater awareness of deepfakes are more capable of identifying such content.

Table 4.11. Effect of Detection Ability and Deepfake Awareness on Reporting Behavior (Paths *b* and *c'*)

Predictor	B	SE	t	p	95% Confidence Interval	
					Lower Limit	Upper Limit
Constant	.915	.737	1.241	.220	-.561	2.391
Awareness (X)	1.307	.337	3.881	<.001	.632	1.981
Detection (M)	1.750	.397	4.413	<.001	.956	2.544

On the second regression model, the effect of both deepfake awareness and detection ability on reporting behaviour. The table showed that both awareness and detection were identified to be significant positive predictors of reporting behaviour. Even after controlling for detection ability, awareness remained a strong predictor (B = 1.307, p < .001), which indicates a direct effect. Also, detection ability has a strong and statistically significant effect on reporting behaviour (B = 1.750, p < .001). This demonstrates that students who have higher awareness of deepfakes and better

detection skills are more likely to report such content. Notably, detection ability appears to be a substantial factor influencing reporting behavior.

Table 4.12. Total Effect of Awareness on Reporting Behavior (Path *c*)

Predictor	B	SE	t	p	95% Confidence Interval	
					Lower Limit	Upper Limit
Constant	1.362	.838	1.625	.110	-.316	3.040
Awareness	1.692	.373	4.532	<.001	.945	2.439

The total effect was evaluated by regressing reporting behavior solely on awareness, without accounting for detection ability in the model. The table indicates that awareness has a statistically significant effect on reporting behavior (B = 1.692, SE = 0.373, t(58) = 4.532, p <.001). This may imply that higher awareness is closely linked to an increased likelihood of reporting deepfake content, before considering any mediating factors.

Table 4.13. Indirect Effect of Awareness on Reporting Behavior Through Detection Ability (*a x b*)

Mediator	Indirect Effect	Boot SE	Boot LLCI	Boot ULCI
Detection	.385	.220	-.014	.851

Bootstrapping was employees to estimate the indirect effect of awareness on reporting behaviour through detection ability. As the 95% bootstrapped confidence interval includes zero ([-.015, .851]), it is evident that the indirect effect (B = 0.385) was not statistically significant. Although the point estimate indicates a positive mediation effect, the lack of statistical significance suggests that there is insufficient evidence to validate that detection ability mediates the relationship between awareness and reporting behaviour at the 95% confidence level.

Moderation Analysis

Similar to the mediation analysis, PROCESS Macro was used to conduct moderation analysis. This is to test whether ethical position (W) moderates the

relationship between deepfake awareness (X) and reporting behaviour (Y) among students through Model 1 (Moderation Model).

Table 4.14. Model Summary

R	R Square	MSE	F	df1	df2	p
.621	.385	.545	11.690	3.000	56.000	.000

The overall model was found to be statistically significant ($F(3,56) = 11.69$, $p < .001$). An $R^2 = .385$ indicates that the combined effect of deepfake awareness, ethical position, and their interaction explains approximately 38.5% of the variance in reporting behaviour.

Table 4.15. Key Coefficients

Predictor	Coefficient (b)	SE	t	p	95% Confidence Interval	
					Lower Limit	Upper Limit
Constant	-16.109	7.108	-2.266	.027	-30.348	-1.869
Awareness (X)	8.818	3.303	2.670	.010	2.201	15.435
Ethical Position (W)	5.600	2.405	2.328	.024	0.782	10.417
Interaction (X × W)	-2.292	1.133	-2.022	.048	-4.562	-0.021

The significant interaction term ($b = -2.292$, $p = .048$) indicates that the effect of deepfake awareness on reporting behaviour varies based on the level of ethical position.

Table 4.16. Conditional Effects of Deepfake Awareness at Different Levels of Ethical Position

Ethical Position (W)	Effect of Awareness on Reporting	SE	t	p	95% Confidence Interval	
					Lower Limit	Upper Limit
Low (M - 1 SD)	2.809	.499	5.629	.000	1.810	3.809
Mean (M)	1.944	.411	4.728	.000	1.120	2.767
High (M + 1 SD)	1.078	.675	1.597	.116	-.274	2.430

The table shows that awareness serves as a strong predictor of reporting behaviour at lower levels of ethical position. At moderate levels, the relationship remains statistically significant and positive. However, at higher levels of ethical position, the effect of awareness on reporting behaviour diminishes and loses statistical significance ($p = .116$). The findings confirm the presence of a moderation effect, wherein ethical position significantly influences the strength of the relationship between awareness and reporting behaviour. Notably, while higher deepfake awareness generally leads to a higher likelihood of reporting, this effect becomes weaker as levels of ethical position increase. This may imply that students with stronger ethical beliefs may depend more on their moral principles over awareness when deciding whether to report deepfake content. On the other hand, students with lower ethical positioning may rely more on their level of awareness to take action.

Analysis and Discussion

In this chapter, the quantitative results will be analyzed, and implications will be discussed based on the research objectives. The findings are evaluated in light of existing literature on the variables such as deepfake awareness, detection ability, ethical position, and reporting behavior.

4.1. Influence of Deepfake Awareness on Reporting Behavior Mediated by Detection Ability

To address the first research objective, the study investigated whether individuals' detection ability mediates the relationship between deepfake awareness and reporting behavior. The mediation analysis demonstrated that deepfake awareness is a significant predictor of detection ability ($p = .045$), which in turn strongly influences reporting behavior ($p < .001$). Although the total and direct effects of awareness on reporting showed significance, the indirect effect through detection ability did not reach statistical significance (95% CI = [-.014, .851]). This indicates a partial but inconclusive mediation. In short, detection ability plays a role, but its mediating effect remains uncertain. Being aware solely of deepfakes does not fully compel students to report that kind of content. Rather, their confidence and ability to detect deepfakes strengthen this relationship. While awareness raises concern among students, detection empowers action. This implication aligns with the ABC Model of Attitude, which is a framework in psychology that characterizes 3 components of attitudes (McLeod, 2023). The Affective component involves the person's emotions about the attitude object. The behavioural component refers to how an individual's attitude influences their behaviour and actions. Cognitive component refers to a person's knowledge and beliefs about a particular attitude object. Deepfake awareness represents the cognitive component, since it involves the knowledge and associated risks of deepfakes. This awareness usually results in emotional fear or concern (the affective component), notably when students recognize the possible harm of deepfakes. However, reporting behaviour (the behavioural component) only materializes when students feel confident and capable of their detection ability, making it a critical enabler. In other words, mediation reveals that attitudes convert into actions more effectively when all three components are aligned, such as beliefs, emotions, and confidence in detection, which also aligns with the principle of behavioural consistency. While detection did not statistically mediate the entire relationship, it plays an essential role in translating knowledge into behavior. Collard (2024) highlights the increasing sophistication of deepfakes and advocates for a zero-trust mindset, which emphasizes the importance of ethical flexibility and media literacy. Additionally, overconfidence in one's ability to identify deepfakes can lead to misjudgments. This overconfidence may hinder the effective conversion of awareness into accurate detection and subsequent reporting. The Signal Detection Theory (SDT)

offers a useful framework for understanding this pathway. SDT postulates that an individual's detection performance depends on both their sensitivity to the signal (deepfake) and the decision criterion set by them. In this context, heightened awareness may enhance sensitivity, but if the decision criterion is not properly calibrated, potentially influenced by overconfidence, detection accuracy may not improve, ultimately affecting reporting behaviour.

4.2 Influence of Deepfake Awareness on Reporting Behaviour Moderated by Detection Ability

To address the second research objective, the study determine whether individuals' ethical position moderates the relationship between deepfake awareness and reporting behavior. The moderation analysis confirmed a significant interaction effect ($p = .048$), which demonstrates that the relationship between awareness and reporting varies based on the ethical positions of individuals. Specifically, awareness had a stronger impact on reporting among individuals with lower ethical idealism ($B = 2.809$), whereas at higher idealism, the effect weakened and became statistically insignificant ($B = 1.078$, $p = .116$). This pattern aligns with Forsyth's Ethics Position Theory (1980), which posits that moral judgements of individuals are influenced by their levels of relativism and idealism. High-idealism students, such as absolutists and situationists, tend to make decisions based on strict moral codes and thus may not act or be influenced by awareness alone unless the deepfake content clearly violates their ethical standards or principles. Conversely, those with low-idealism students, such as subjectivists and exceptionists, depend more on their personal values, empathy, and situational factors, which makes them more likely to allow awareness to strongly influence their reporting behavior. The findings are consistent with Gomez's (2024) study, which mentioned that the ethical responses to deepfakes rely not only on consequences, but also on the moral perspectives individuals adopt. From a consequentialist perspective, deepfakes are unethical when they cause harm or deceive people, which aligns with how subjectivists and exceptionists may act towards that content. From a deontological perspective, like may refrain from engaging unless moral duties are violated, which aligns with absolutists and situationists. Furthermore, Gomez emphasized how the absence of regulation and ethical ambiguity surrounding deepfakes, like falsified evidence or non-consensual deepfake pornography, demands that individuals and engineers exercise proactive moral judgement. These nuances reinforce the study's findings that ethical position moderates not only perceptions of deepfakes, but also whether they feel compelled to act upon it. While Gomez stressed ethical ambiguity in deepfakes, the moderation found in this study is better explained by Information Manipulation Theory (IMT), which explains that deceptive content works by subtly breaking conversational norms such as quantity, quality, manner, and relevance

(McCornack, 2015). According to the theory, people assess these violations not solely cognitively but also through their ethical or moral lenses. This further reinforces the notion that responses to deepfakes are influenced by both ethical position and situational understanding. Ultimately, these findings suggest the importance of awareness campaigns that integrate both ethical considerations and cognitive understanding, ensuring individuals' moral values are effectively aligned with their online reporting practices.

4.3 Direct Effect of Deepfake Awareness on Reporting Behavior

To address the third research objective, the study examined whether there is a direct effect on the relationship between deepfake awareness and reporting behavior. The linear regression analysis revealed a significant positive relationship between deepfake awareness and reporting behavior ($B = 1.692$, $p < .001$), which explains 26.2% of the variance. This highlights the essential role of awareness in motivating students to report deepfakes. This supports the Theory of Planned Behavior (Ajzen, 2019), which suggests that attitudes and perceived behavioral control are important in shaping people's intentions and actions. However, the unexplained 73.8% variance suggests that some psychological and contextual factors are likely to influence reporting behavior. One potential factor is confidence in detection ability. While a lot of people are aware of deepfakes, their confidence in detecting them is relatively low. According to the study of Sippy et al. (2024), it was found that although 90.4% of the participants were concerned about the emergence of deepfakes, most were not confident in their ability to detect them. This illustrates the gap between awareness and reporting action, which stresses the need for interventions to enhance detection skills. A study by Cochran and Napshin (2021) reinforces this gap by revealing that while deepfake awareness and concern are widespread, factors like individual responsibility can reduce the sense of urgency for action. Interestingly, even with the doubts in current technological solutions, there remains a strong public support for platform-level interventions, including deepfake detection systems. This strengthens the idea that awareness alone is insufficient; effective reporting behavior is more likely when paired with platform accountability, detection skills, and ethical engagement.

Conclusions, Recommendations and Limitations

This study investigates how deepfake awareness, detection ability, and ethical position directly and indirectly shape students' willingness to report deceptive media, like deepfakes, in Metro Manila. It was revealed that greater awareness significantly increased the intention to report deepfakes, with detection ability partially mediating this relationship and ethical position moderating it. Students with stronger detection abilities and lower idealism were more likely to recognize and report deepfakes. The findings confirmed that these variables collectively influence a student's likelihood to take action against harmful synthetic media. It highlights how crucial it is to foster both ethical reasoning and media literacy, especially in the modern era, where individuals' daily lives are increasingly affected by misinformation, social media, artificial intelligence, and synthetic content. Ultimately, this study sheds light on the behavioural responses of students and the interventions needed to minimize the societal impact. It emphasizes the urgency of building stronger technological, educational, and legal infrastructures to address the issues posed by deepfakes or similar content.

Recommendations

1. For Educational Institutions
    - Educational institutions should incorporate modules on misinformation and synthetic content on digital literacy subject to tackle deepfakes and spread awareness among students. These modules should not only include theoretical discussions, but also integrate interactive exercises, like practicing their identification of real vs fake media. This improves their detection ability and critical thinking. Teaching students about deepfakes risks, manipulation tactics, and ethical concerns will help close the gap between awareness and detection that was identified in this study, which prepares them to navigate today's complex and ever evolving digital landscape.
    - Beyond classrooms, educational programs and seminars can be organized and taught by AI experts or cybersecurity professionals to teach about the basics of deepfake creation, potential risks, and manipulation indicators such as unnatural facial movements or odd audio inconsistencies.
    - In addition, institutions may promote responsible digital consumption and verification tools by curating resource collection through workshops and physical or digital references.
2. For Social Media Platforms
    - Social media platforms should improve their reporting systems by incorporating AI-powered deepfake detectors and simplifying reporting

process, which makes it accessible and quicker for users. Moreover, platforms must establish immediate reporting mechanisms that prioritize review and takedown of harmful and malicious deepfake posts as soon as it's flagged. Once a ticket or report is submitted, clear and transparent protocols should be in place to evaluate the violation swiftly, with appropriate sanctions including account suspension, post removal, or stricter penalties for digital creators of non-consensual or harmful deepfake media. Social media companies should treat deepfake-related violations rigorously by classifying them under particular community guidelines that are equivalent to harassment, misinformation, and privacy violations. This makes the digital space for everyone safer and trustworthy.

3. For Policymakers
    - Policymakers in the Philippines should formulate comprehensive laws addressing deepfakes, not only imposing penalties on creators of harmful deepfakes but also on protecting victims' privacy rights. Additionally, there must be laws that safeguards individuals from AI-powered defamation, identity theft, and harassment, especially the growing prevalence of social media and artificial intelligence. Victims of such incidents must have clear legal pathways to seek justice. This recommendation is echoed in a news article by Felicia (2024), wherein it emphasizes the lack of deepfake laws in the Philippines.
    - One of the recommendations is the implementation of mandatory labelling of AI-generated content, wherein a visible disclaimer must be indicated on posts in order for the users to distinguish synthetic from authentic media. In that way, the spread of misinformation will be reduced.
    - Government agencies must launch nationwide educational initiatives to teach Filipinos how to detect manipulated media and protect themselves from the risks associated with deepfakes. Digital literacy programs can be integrated not only in schools, but also workplaces and community centres to reach a wider audience.
4. For Future Researchers
    - Future studies should examine thoroughly the role of perceptual cues (e.g., facial distortions, unnatural movements, video quality, audio inconsistencies) that help or mislead individuals when detecting deepfakes. With this understanding about these signs, this allows for effective educational interventions that enhance awareness and detection abilities.
    - Other than perceptual cues, overconfidence bias should be further studied, given its significant potential to distort detection judgements, which makes individuals more vulnerable to deepfakes.

- Researchers must also consider utilizing Structural Equation Modelling (SEM) with the variables used in this study for future investigations. SEM would enable for more sophisticated and advanced causal modelling of the relationships between deepfake awareness, detection ability, ethical position, and reporting behaviour. This offers deeper insights than the simple regression, mediation, and moderation methods applied in the study.

Limitations

One of the main limitations of this study was the small sample size, wherein it only gathered data from 60 respondents, thus limiting the generalizability of the findings. A larger sample size will strengthen the statistical reliability and validity of the results, which enables for robust interpretations in future studies. The geographical context is only limited as well to students in Metro Manila, which limits the diversity, given that snowball sampling method was used. The cross-sectional nature of the study only captured attitudes and behaviour at a single point in time and could not account for changes in detection ability and perception over time. Longitudinal study would help understand whether exposure and education improve detection abilities of students. Additionally, the study was self-reported, meaning there is no outside interference, which makes the findings susceptible to bias including inaccuracies in self-assessment and social desirability bias. Since detection ability was partly measures through video-based tasks, it posed challenges in terms of variations in devices (e.g., phones, laptops, tablets). These variations may have disparities in screen and size resolution, affecting detection accuracy, which introduced an uncontrolled variable in the study. In Lastly, while the study mentioned psychological and social factors, it did not deeply examine the cultural influences, which could provide more valuable insights.

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Appendices

Appendix B: Ethics Compliance

Copy of Research Ethics Form

Informed Consent

Before participating in this study, it is important that you understand why this research is being done and what are the things involved. Kindly take time to read the materials below. This section explains the purpose of the study and how your personal information will be collected, used, and protected in accordance with the Data Privacy Act of 2012 (RA 10173).

If you do not understand any part of this form, please contact the researcher for clarification. Likewise, if you have any questions or concerns about your privacy rights or other matters, please feel free to contact me with the details provided below. I sincerely appreciate considering participating in my data collection.

Amanda A. Ang
23-1-0003@sgen.edu.ph

Purpose of study

This study aims to explore the relationship between deepfake awareness, ethical position, detection ability, and reporting behavior among students in Metro Manila. This research is being conducted as part of my research requirement for my course in Pearson HND in Computing, Data Analytics.

Deepfake refers to synthetic media created using deep learning techniques, which manipulate videos, images, or audio to make them appear real (Silva et al., 2022). While deepfake technology offers potential benefits in fields like entertainment and education (Gupta, 2023), it raises significant ethical concerns, particularly regarding disinformation and privacy (Fallis, 2020). The widespread use of deepfakes on social media has led to reputational damage, psychological trauma, and financial loss for its victims (Sharma et al., 2023).

Specifically, I will explore how detection ability mediates the relationship between awareness and reporting behavior, and how an individual's ethical position moderates this connection. By addressing this gap, the study will contribute to the understanding of how public awareness, ethical responsibility, and detection skills influence actions against deepfake content and misinformation.

The findings of this study will be valuable for enhancing digital literacy and informing educational campaigns to raise awareness of deepfakes, particularly among students in Metro Manila, who are frequent social media users. The research will help develop strategies for encouraging responsible online behavior and mitigating the negative impacts of deepfake technology.

☐ I have read and understood this section.

Voluntary Participation

Participation in this study is entirely voluntary. At any point, you may choose to stop, and withdraw your consent at any point during or after the data collection, regardless of the reason/s. In addition, you may opt out of the study if you participate but meet one of the exclusion criteria, or if you suffer any kind of discomfort while answering surveys. Additionally, those who are unable or do not have the capacity to continue will be excluded. You are ensured that by withdrawing consent, your choices will be respected and will not be held against them. Further, this will not impact their relationship with the researcher, nor with Southville International School affiliated with Foreign Universities (SISFU).

☐ I have read and understood this section.

Procedure & Duration

Your participation in this research study will not affect the outcome of the results if you are not willing to participate. Upon agreeing to participate, you will be asked to complete this online survey through Google Forms that will take approximately 20-40 minutes of your time. Please ensure you have enough time to spare to answer thoughtfully. The survey asks for your responses to this consent form, a demographic questionnaire, Deepfake Risk and Opportunity Perception Questionnaire, The Ethics Position Questionnaire–Short Form (EPQ), Perceived Deepfake Trustworthiness Questionnaire (PDTQ), and The Theory of Planned Behavior Questionnaire (TPBQ).

Below are the brief descriptions of the main instruments that will be used:

Deepfake Risk and Opportunity Perception Questionnaire (Bitton et al., 2021), assesses participants' awareness of the risks and opportunities of deepfake technology. It measures concerns about deepfake-related threats, as well as perceptions of its potential benefits in fields like entertainment, education, and creative industries.

The Forsyth's Ethics Position Questionnaire (EPQ) – Short Form (Forsyth, 2011), is a survey based on Forsyth's Ethics Position Theory (EPT), which examines two dimensions of moral values: relativism and idealism. It measures individuals' tendency to either reject or uphold universal moral principles (relativism) and their focus on minimizing harm versus prioritizing consequences (idealism).

Perceived Deepfake Trustworthiness Questionnaire (PDTQ) (Plohl et al.,2024), assesses participants' perceptions of the trustworthiness of deepfake videos in terms of content and presentation. It evaluates aspects such as the plausibility of information, credibility of the source, and visual or auditory inconsistencies in the video.

The Theory of Planned Behavior Questionnaire (TPBQ) (Ajzen, 2019), assesses the factors influencing behavior, including attitudes, subjective norms, perceived behavioral control, and intentions. It uses tailored items to measure beliefs, motivations, and background variables, providing insights into the predictors of a specific behavior.

☐ I have read and understood this section.

Risks

The risks of participation are minimal but may include discomfort when recalling experiences related to deepfake content or ethical concerns surrounding misinformation. You might find certain questions or topics distressing, particularly if they involve personal experiences with social media or privacy. Should you feel uncomfortable, you are free to skip any questions or withdraw from the study entirely at any time by exiting the survey page or closing your browser. Your participation is voluntary, and your decision to stop will not affect your relationship with the researcher. There will be no negative consequences for withdrawing from the study.

☐ I have read and understood this section.

Benefits

While there are no direct benefits to your participation, your involvement in this study may offer the opportunity for self-reflection on your awareness and attitudes toward deepfakes, misinformation, and ethical issues related to digital media. The findings from this research may:

1. Contribute to a better understanding of how awareness of deepfakes, ethical positions, and detection abilities impact an individual's likelihood of reporting fake content, thus addressing gaps in the literature surrounding deepfakes.
2. Inform digital literacy programs and awareness campaigns aimed at educating the public, particularly students, about the ethical implications of deepfake technology and the importance of responsible media consumption.
3. Help shape the development of policies and tools for combating misinformation on social media platforms, contributing to stronger regulations for protecting individuals' rights and privacy in the digital age.

☐ I have read and understood this section.

Confidentiality

All personal information collected will be held with the utmost confidentiality. Your identity as a participant will be anonymized. The personal data and responses will be stored securely. Only relevant information will be collected and the researcher is prohibited from sharing information to anyone. Only the researcher and you, the participant, will have access to the data, provided such access will solely be used in this research study. All of these are bound by the Data Privacy Act of 2012.

☐ I have read and understood this section.

Data Usage

Your participation in this study requires that you voluntarily grant the researcher your permission to use your information solely for data analysis and academic consultation purposes in fulfillment of the researcher's academic requirement. The screening tool and measuring tools will be answerable via Google Forms, which is end-to-end encrypted. Hence, third parties will not be able to access the data provided by the participants and only the intended recipient can decrypt it. After the data is interpreted, and upon the completion of the research paper, the researcher will destroy all the information gathered. The results of this study may only be requested physically at the researcher's institution, provided that the researchers finished all write-ups within the given timeframe. If the researcher wants to use your information for something outside of this specific study, they'll need to ask for your permission again with a separate consent form.

☐ I have read and understood this section.

Certificate of Consent:

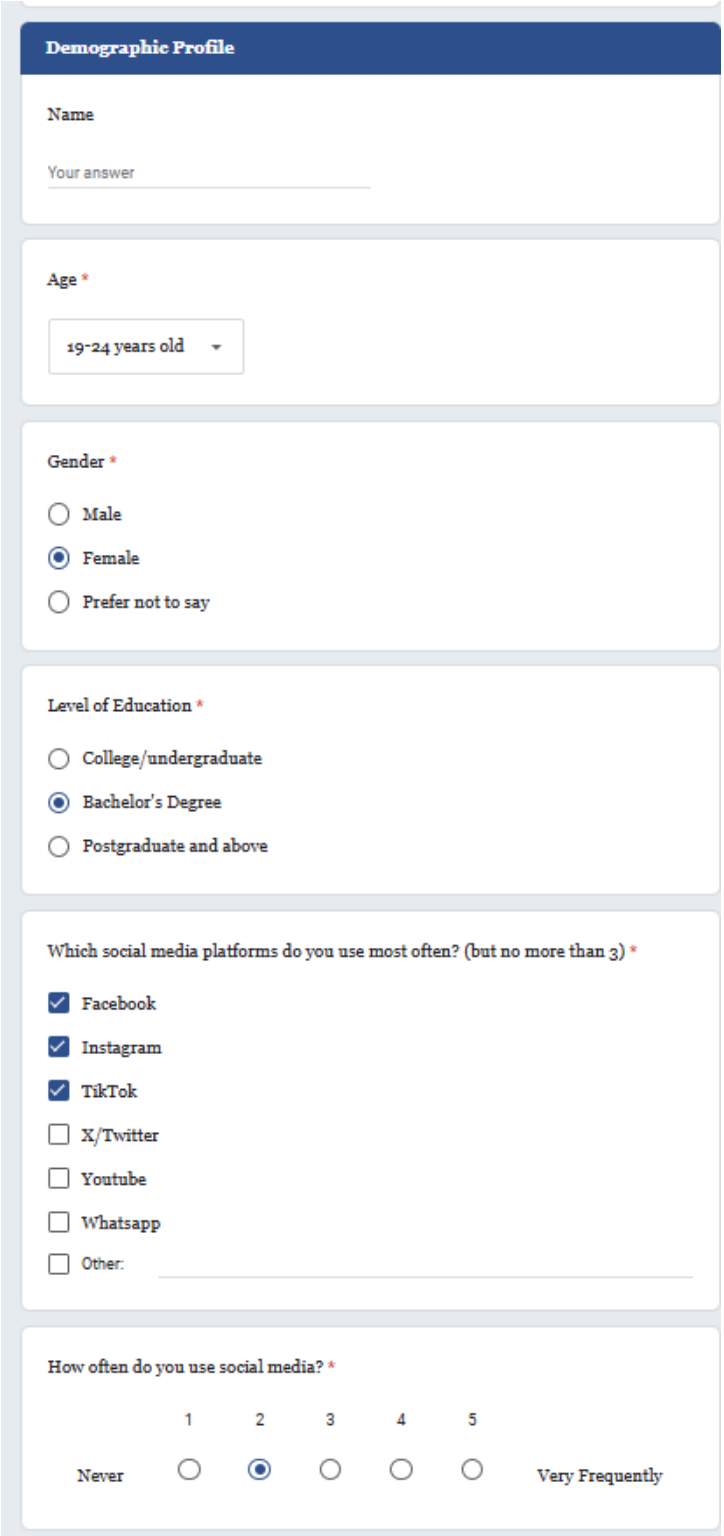
I have read and fully understood the information and the privacy notice provided in this consent form. I have been given an opportunity to ask questions and any questions I have asked have been answered to my satisfaction.

I consent to the use of my personal records, survey responses, and findings collected during this study for educational purposes and academic presentation only, provided that my identity will remain confidential.

◯ Yes, I consent voluntarily, out of my own free will, to participate in this research study.

◯ No, I decline participating in this study.

Appendix C: Data Collection Instruments



Demographic Profile

Name

Your answer

Age *

19-24 years old

Gender *

- Male
- Female
- Prefer not to say

Level of Education *

- College/undergraduate
- Bachelor's Degree
- Postgraduate and above

Which social media platforms do you use most often? (but no more than 3) *

- [x] Facebook
- [x] Instagram
- [x] TikTok
- [ ] X/Twitter
- [ ] Youtube
- [ ] Whatsapp
- [ ] Other:

How often do you use social media? *

	1	2	3	4	5	
Never	○	◉	○	○	○	Very Frequently

How comfortable are you using digital tools or technology? *

	1	2	3	4	5	
Not comfortable at all	○	○	○	◉	○	Very comfortable

How often do you use AI-powered tools or applications (e.g., ChatGPT, voice assistants, image generators)? *

	1	2	3	4	5	
Never	○	○	○	◉	○	Very Frequently

Have you ever encountered deepfake in social media? *

- ◉ Yes
- ○ No

If yes, what type of deepfake content have you encountered? (Select all that apply)

- ☑ Videos (e.g., manipulated speeches, fake interviews)
- ☑ Images (e.g., altered photos of people)
- ☐ Audio (e.g., voice cloning, fake calls)
- ☐ Other: _____

How often do you encounter potentially manipulated or fake content on social media? *

	1	2	3	4	5	
Never	○	○	○	◉	○	Very Frequently

Deepfake Awareness

Please indicate your level of agreement with the following statements (15 items).

1 = Completely or rather disagree
2 = Neither nor
3 = Completely or agree

I am concerned about falling for a deepfake video on social media platforms

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I am concerned about becoming a victim of a harmful deepfake (e.g., voice cloning, pornographic deepfakes, or scams)

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I am concerned that deepfakes can be used to influence public opinion or sway election campaigns

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I worry that, in the future, I won't be able to trust any video content due to the increasing use of deepfake technology.

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I worry that deepfakes will negatively affect public trust and contribute to fake news.

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I am concerned that almost anyone with basic tools can create deepfakes, including videos and audio clips of public figures.

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I believe that personal data manipulation will become easier with deepfake technology.

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I am concerned that I will need to take extra steps to protect my personal data.

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I think deepfakes can also be used for beneficial purposes.

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I believe deepfakes provide me with benefits (e.g., by creating funny videos with TikTok filters or voice cloning apps).

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I think there are advantages in the fact that anyone can create deepfakes (e.g., through easily accessible apps like Reface or Zao).

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I believe deepfakes provide benefits to the film industry (e.g., digitally resurrecting deceased actors in movies).

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I believe deepfakes offer advantages for the fashion industry (e.g., digital avatars for modeling).

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I believe deepfakes foster creativity in media content production (e.g., enabling influencers to produce unique videos).

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

I believe deepfakes can enhance user experience on social media platforms (e.g., personalized filters, voice transformations, and video remixes).

	1	2	3	
Completely or rather disagree	○	○	○	Completely or rather agree

Ethical Position

Please indicate if you agree or disagree with the following items. Each represents a commonly held opinion and there are no right or wrong answers (15 items).

1 = Strongly disagree
2 = Disagree
3 = Neither agree nor disagree
4 = Agree
5= Strongly agree

A person should make certain that their actions never intentionally harm another even to a small degree.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

The existence of potential harm to others is always wrong, irrespective of the benefits to be gained.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

One should never psychologically or physically harm another person.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

One should not perform an action which might in any way threaten the dignity and welfare of another individual.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

If an action could harm an innocent other, then it should not be done.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

What is ethical varies from one situation and society to another.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Moral standards should be seen as being individualistic; what one person considers to be moral may be judged to be immoral by another person.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Questions of what is ethical for everyone can never be resolved since what is moral or immoral is up to the individual.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Moral standards are simply personal rules that indicate how a person should behave and are not applied in making judgments of others.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Ethical considerations in interpersonal relations are so complex that individuals should be allowed to formulate their own individual codes.

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Is it ethical to use deepfakes for political satire, even if it risks misleading the public or manipulating emotions?

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Do you believe it is ethical to use deepfake technology for entertainment purposes even if it might blur the line between reality and fiction?

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Is it ethical to create deepfake content that might cause distress to individuals, even if it serves artistic or creative purposes?

	1	2	3	4	5	
Strongly Disagree	○	○	○	○	○	Strongly Agree

Select ONE of the following statements that you most agree with: *

- ○ The existence of potential harm to others through deepfakes is always wrong, irrespective of the benefits to be gained.
- ○ Sometimes, even the best use of deepfake technology may cause harm to others.

Select ONE of the following statements that you most agree with: *

- ○ Moral standards regarding the use of deepfakes should be seen as individualistic; what one person considers moral may be judged immoral by another person.
- ○ Moral principles and standards regarding the ethical use of deepfake technology define what is morally right and what is morally wrong.

Detection Ability

Please watch the following 5 video samples and complete the following tasks for each one (10 items for each video sample).

1 = Strongly Disagree
2 = Disagree
3 = Somewhat Disagree
4 = Neutral
5 = Somewhat Agree
6 = Agree
7 = Strongly Agree

Disclaimer:
The videos presented in this section include a mix of real and fake content, gathered from various social media platforms. All videos are used solely for academic purposes to evaluate detection ability. The inclusion of any video does not reflect endorsement or disapproval of its content, and the message or content of the videos is not related to the study.

Video #1



If the video attached is unavailable, please click the link above.

Is this video real or fake? *

- ◯ Real
- ◯ Fake

How confident are you that the video is real/fake? *

	1	2	3	4	5	
Not confident at all	◯	◯	◯	◯	◯	Very confident

The mouth movements of the person in the video did not completely match the sound.

	1	2	3	4	5	6	7	
Strongly Disagree	○	○	○	○	○	○	○	Strongly Agree

The facial features of the person in the video changed during the video.

	1	2	3	4	5	6	7	
Strongly Disagree	○	○	○	○	○	○	○	Strongly Agree

The person's gestures in the video did not seem natural.

	1	2	3	4	5	6	7	
Strongly Disagree	○	○	○	○	○	○	○	Strongly Agree

The person's gestures in the video did not seem natural.

	1	2	3	4	5	6	7	
Strongly Disagree	○	○	○	○	○	○	○	Strongly Agree

The video quality was inconsistent.

	1	2	3	4	5	6	7	
Strongly Disagree	○	○	○	○	○	○	○	Strongly Agree

The face of the person in the video (or parts of it) was distorted.

	1	2	3	4	5	6	7	
Strongly Disagree	○	○	○	○	○	○	○	Strongly Agree

The mouth of the person in the video was moving strangely.

	1	2	3	4	5	6	7	
Strongly Disagree	○	○	○	○	○	○	○	Strongly Agree

The presented information seemed questionable.

	1	2	3	4	5	6	7	
Strongly Disagree	○	○	○	○	○	○	○	Strongly Agree

Reporting Behavior

Reporting a deepfake would help reduce the spread of misleading information. *

	1	2	3	4	5	6	7	
Unlikely	○	○	○	○	○	○	○	Likely

I believe it is important to report deepfake videos to stop the spread of misinformation *

	1	2	3	4	5	6	7	
Disagree	○	○	○	○	○	○	○	Agree

People I care about think I should report deepfake videos if I come across them. *

	1	2	3	4	5	6	7	
Disagree	○	○	○	○	○	○	○	Agree

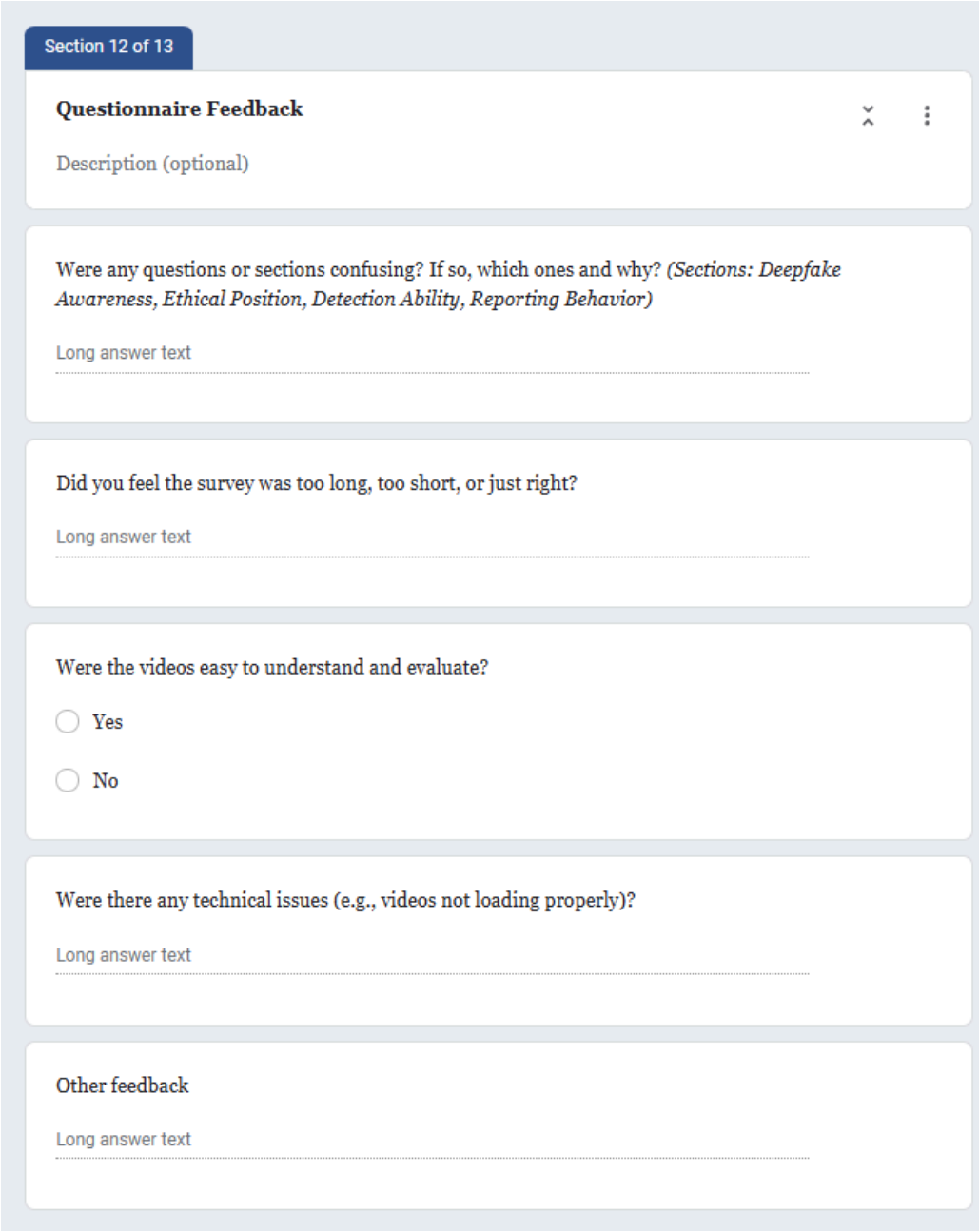
If I come across a deepfake, I intend to report it. *

	1	2	3	4	5	6	7	
Not at all	○	○	○	○	○	○	○	Very much

If I saw a deepfake video of a celebrity, how likely are you to report it? *

	1	2	3	4	5	6	7	
Not at all	○	○	○	○	○	○	○	Very much

Pilot Testing Questionnaire Feedback Questions



Appendix D: Actual Data Summaries or Transcripts

Reliability

Scale: ALL VARIABLES

Case Processing Summary

		N	%
Cases	Valid	10	100.0
	Excluded[a]	0	.0
	Total	10	100.0

a. Listwise deletion based on all variables in the procedure.

Initial Deepfake Awareness Reliability Statistics

Cronbach's Alpha	N of Items
.803	20

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
falling for a deepfake video	47.30	32.456	.000	.805
becoming a victim of a harmful deepfake	47.40	30.933	.404	.796
influence public opinion	47.30	32.456	.000	.805
won't be able to trust any video content	47.30	32.456	.000	.805
anyone with basic tools can create deepfakes	47.50	32.056	.047	.808
personal data manipulation will become easier	47.30	32.456	.000	.805
extra steps to protect my personal data	47.40	31.156	.340	.798
rising prevalence	47.50	33.611	-.212	.826
used for beneficial purposes	47.90	24.989	.801	.760
deepfakes provide me with benefits	47.80	26.178	.799	.765

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
anyone can create deepfakes	48.00	25.111	.808	.760
film industry	48.50	26.500	.657	.774
art industry	48.90	29.656	.303	.799
fashion industry	48.00	27.778	.461	.789
video game industry	47.70	28.233	.502	.786
creativity in media content production	47.90	25.211	.772	.763
education and training	48.80	30.178	.229	.804
social media platforms	48.20	25.289	.618	.775
blurring the line between reality and fiction	47.60	34.044	-.260	.831
negatively affect public trust	47.40	31.156	.340	.798

Reliability

Case Processing Summary

		N	%
Cases	Valid	10	100.0
	Excluded[a]	0	.0
	Total	10	100.0

a. Listwise deletion based on all variables in the procedure.

Final Deepfake Awareness Reliability Statistics

Cronbach's Alpha	N of Items
.855	18

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
falling for a deepfake video	41.80	35.511	.000	.858
becoming a victim of a harmful deepfake	41.90	33.878	.417	.851
influence public opinion	41.80	35.511	.000	.858
won't be able to trust any video content	41.80	35.511	.000	.858
anyone with basic tools can create deepfakes	42.00	34.889	.089	.859
personal data manipulation will become easier	41.80	35.511	.000	.858
extra steps to protect my personal data	41.90	34.100	.355	.852
used for beneficial purposes	42.40	27.378	.841	.824
deepfakes provide me with benefits	42.30	28.678	.836	.827
anyone can create deepfakes	42.50	27.833	.806	.827
film industry	43.00	30.000	.566	.842
art industry	43.40	32.489	.318	.855
fashion industry	42.50	30.056	.529	.845
video game industry	42.20	30.844	.538	.844
creativity in media content production	42.40	27.600	.813	.826
education and training	43.30	33.789	.149	.863
social media platforms	42.70	27.344	.690	.835
negatively affect public trust	41.90	34.100	.355	.852

Reliability

Scale: ALL VARIABLES

Case Processing Summary

		N	%
Cases	Valid	10	100.0
	Excluded[a]	0	.0
	Total	10	100.0

a. Listwise deletion based on all variables in the procedure.

Initial Ethical Position Reliability Statistics

Cronbach's Alpha	N of Items
.517	13

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
even to a small degree	37.90	24.100	-.245	.556
irrespective of the benefits	39.10	24.100	-.210	.623
never physically harm	37.80	23.956	-.215	.554
that might threaten the dignity	37.80	23.956	-.215	.554
then it should not be done	37.90	24.100	-.245	.556
one situation and society	39.70	18.456	.450	.432
should be seen as individualistic immoral by another person.]	39.80	17.289	.355	.444
what is moral or immoral	38.80	21.511	.113	.515
simply personal rules	39.00	18.444	.289	.469
own individual codes	39.00	17.111	.341	.449
deepfakes for political satire	40.50	17.167	.671	.377
entertainment purposes	40.00	17.111	.783	.364
artistic or creative purposes	40.30	19.122	.467	.440

Reliability

Reliability

Scale: ALL VARIABLES

Case Processing Summary

		N	%
Cases	Valid	10	100.0
	Excluded[a]	0	.0
	Total	10	100.0

a. Listwise deletion based on all variables in the procedure.

Final Ethical Position Reliability Statistics

Cronbach's Alpha	N of Items
.832	8

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
one situation and society	18.70	25.567	.769	.787
should be seen as individualistic immoral by another person.]	18.80	24.178	.601	.809
what is moral or immoral	17.80	28.844	.487	.822
simply personal rules	18.00	26.000	.505	.822
own individual codes	18.00	24.444	.539	.821
deepfakes for political satire	19.50	28.056	.525	.817
entertainment purposes	19.00	28.222	.584	.812
artistic or creative purposes	19.30	28.011	.617	.809

Reliability

Scale: ALL VARIABLES

Case Processing Summary

		N	%
Cases	Valid	10	100.0
	Excluded[a]	0	.0
	Total	10	100.0

a. Listwise deletion based on all variables in the procedure.

Initial Detection Ability
Reliability Statistics

Cronbach's Alpha	N of Items
.800	12

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
convincing	31.80	152.622	-.561	.868
match the sound	35.10	99.656	.903	.742
plausible	31.90	154.322	-.553	.875
unnatural	34.40	108.711	.544	.776
facial features	34.60	96.267	.871	.738
gesture	34.80	99.511	.901	.741
inconsistent	35.00	96.000	.938	.733
face of the person	34.90	94.767	.939	.731
moving strangely	35.00	99.778	.938	.740
questionable	34.50	89.167	.714	.753
real or fake	36.20	128.400	.340	.800
How confident	33.20	134.178	-.188	.815

Reliability

Scale: ALL VARIABLES

Case Processing Summary

		N	%
Cases	Valid	10	100.0
	Excluded[a]	0	.0
	Total	10	100.0

a. Listwise deletion based on all variables in the procedure.

Final Detection Ability
Reliability Statistics

Cronbach's Alpha	N of Items
.941	9

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
match the sound	21.00	121.333	.875	.928
facial features	20.50	113.611	.962	.922
gesture	20.70	120.011	.910	.926
inconsistent	20.90	116.100	.949	.923
face of the person	20.80	113.511	.986	.920
moving strangely	20.90	121.211	.917	.926
questionable	20.40	100.933	.897	.933
real or fake	22.10	149.433	.578	.949
How confident	19.10	158.322	-.198	.962

Reliability

Scale: ALL VARIABLES

Case Processing Summary

		N	%
Cases	Valid	10	100.0
	Excluded[a]	0	.0
	Total	10	100.0

a. Listwise deletion based on all variables in the procedure.

Initial Reporting Behavior Reliability Statistics

Cronbach's Alpha	N of Items
.208	11

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
reduce the spread	54.30	14.678	.520	.007
protect individuals from harm	55.20	16.178	.010	.237
report deepfakes	55.50	16.722	.009	.230
social circle	54.60	17.378	.142	.169
necessary tools	55.30	21.789	-.376	.412
deepfake detection	54.60	18.933	-.106	.260
stop the spread	54.50	14.722	.259	.077
report deepfake videos	55.30	11.789	.695	-.201[a]
I feel confident	55.10	23.656	-.567	.443
intend to report	54.80	15.067	.325	.064
celebrity	54.80	14.844	.300	.065

a. The value is negative due to a negative average covariance among items. This violates reliability model assumptions. You may want to check item codings.

Reliability

Scale: ALL VARIABLES

Case Processing Summary

		N	%
Cases	Valid	10	100.0
	Excluded[a]	0	.0
	Total	10	100.0

a. Listwise deletion based on all variables in the procedure.

Final Reporting Behavior Reliability Statistics

Cronbach's Alpha	N of Items
.812	5

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
reduce the spread	22.20	11.956	.910	.706
stop the spread	22.40	10.933	.663	.756
report deepfake videos	23.20	13.067	.431	.828
intend to report	22.70	13.789	.412	.827
celebrity	22.70	11.344	.697	.744

Overall Reliability of the Survey

Cronbach's Alpha	N of Items
.883	40

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
falling for a deepfake video	117.30	338.900	.000	.884
becoming a victim of a harmful deepfake	117.40	335.156	.315	.882
influence public opinion	117.30	338.900	.000	.884
won't be able to trust any video content	117.30	338.900	.000	.884
negatively affect public trust	117.40	332.489	.547	.881
anyone with basic tools can create deepfakes	117.50	330.722	.522	.881
personal data manipulation will become easier	117.30	338.900	.000	.884
extra steps to protect my personal data	117.40	332.489	.547	.881
used for beneficial purposes	117.90	325.433	.419	.880
deepfakes provide me with benefits	117.80	331.733	.259	.882
anyone can create deepfakes	118.00	328.222	.335	.881
film industry	118.50	328.944	.326	.881
fashion industry	118.00	328.889	.313	.881
creativity in media content production	117.90	330.989	.235	.882
social media platforms	118.20	322.178	.441	.879
that might threaten the dignity	115.80	347.067	-.430	.888
one situation and society	117.70	313.789	.706	.875

should be seen as individualistic immoral by another person.]	117.80	309.956	.569	.876
what is moral or immoral	116.80	318.844	.637	.877
simply personal rules	117.00	338.222	-.019	.888
own individual codes	117.00	332.000	.095	.887
deepfakes for political satire	118.50	321.833	.492	.879
entertainment purposes	118.00	322.444	.534	.878
artistic or creative purposes	118.30	324.233	.476	.879
match the sound	118.00	292.222	.786	.870
facial features	117.50	285.611	.786	.870
gesture	117.70	295.567	.718	.872
inconsistent	117.90	285.433	.838	.868
face of the person	117.80	285.733	.800	.869
moving strangely	117.90	290.322	.857	.869
questionable	117.40	283.600	.543	.881
Select one	118.40	338.933	-.011	.884
Select one	118.60	344.489	-.325	.887
real or fake	119.10	336.322	.155	.883
How confident	116.10	340.544	-.088	.886
reduce the spread	114.20	330.178	.250	.882
stop the spread	114.40	323.378	.300	.882
report deepfake videos	115.20	314.178	.549	.877
intend to report	114.70	332.900	.123	.885
celebrity	114.70	334.678	.066	.886